

AVD Deployment and Management

John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com

Shawn Meyer – Principal CSA
Microsoft Federal
shawn.meyer@microsoft.com



Topics



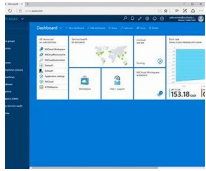
Prerequisites

Deployment

Manage

Secure

Prerequisites



An Azure account with an active subscription

Azure Virtual Desktop configuration, monitoring and management happens in the Azure Portal.

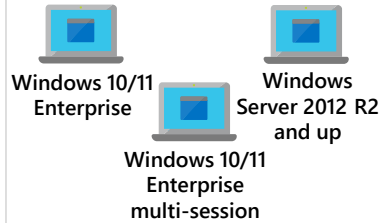
Admins need an account with the necessary privileges to start an Azure Virtual Desktop implementation



An identity provider

Identity providers allow for granular control of user access to Azure Virtual Desktop resources.

Microsoft Entra ID and Active Directory provide identity management for Azure Virtual Desktop



A supported operating system

Azure Virtual Desktop resources are delivered through Virtual Machines running Windows operating systems. You can choose from a broad set of available OS images or create your own



Azure ExpressRoute

Network connectivity

Users can connect to Azure Virtual Desktop from virtually anywhere. Customers can also use Azure ExpressRoute to link on-prem infrastructure to Azure Virtual Desktop services



Appropriate licenses

Many customers are already entitled to access Azure Virtual Desktop resources through existing eligible licenses



Azure Virtual Desktop

A Remote Desktop client

Users can access Azure Virtual Desktop resources through a variety of remote desktop clients

Recommended identity setup for hybrid organizations

Identities are managed via Windows Server Active Directory

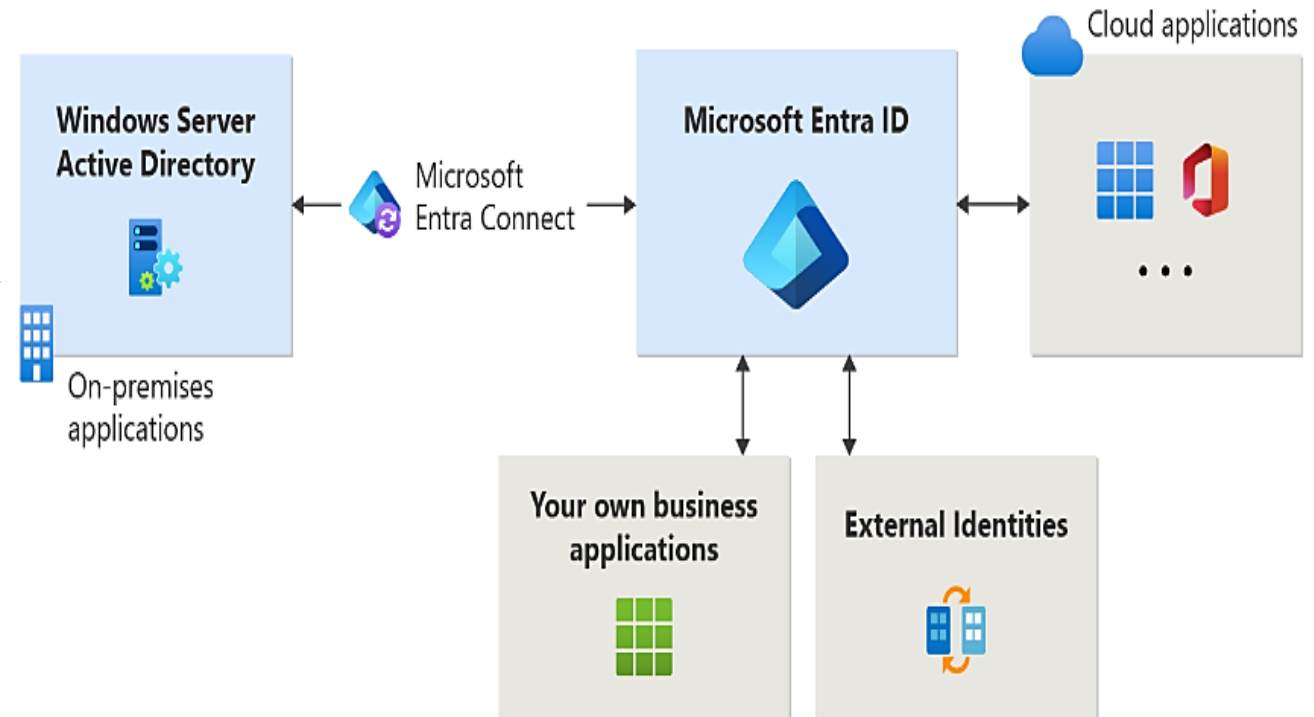


Microsoft Entra ID



Windows Server AD on-prem connected to Azure

- ExpressRoute or site-to-site VPN to Azure
- Microsoft Entra Connect to synchronize identities



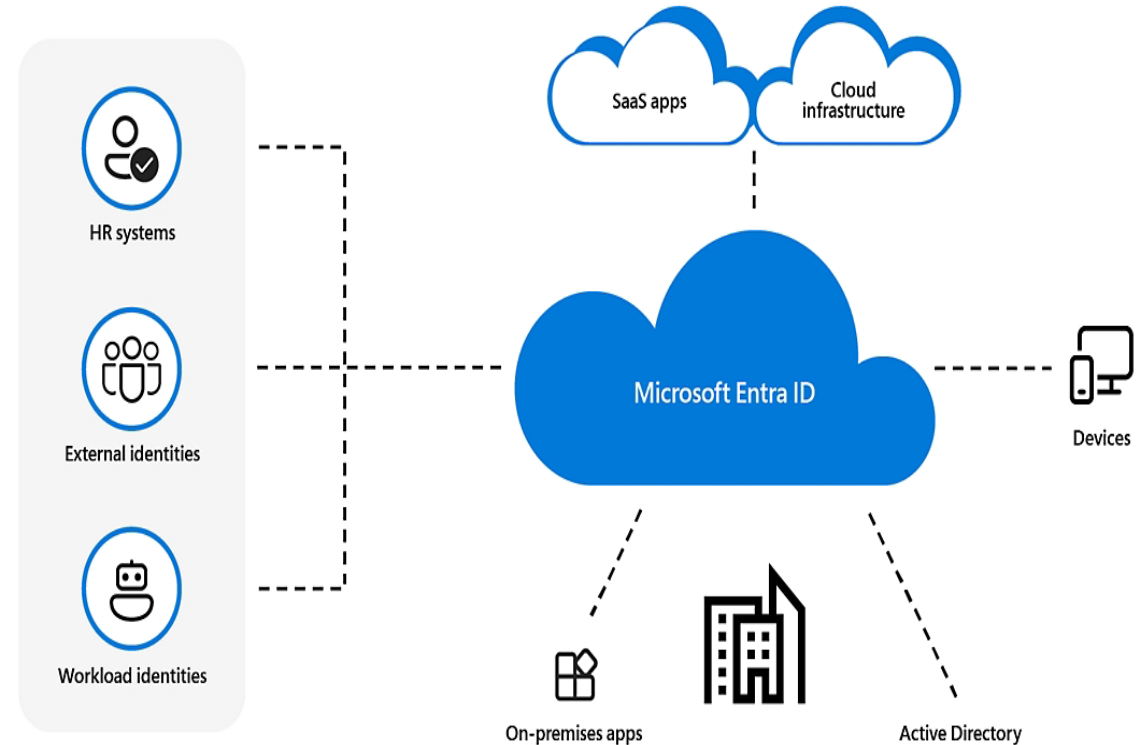
Recommended Identity setup for cloud-based organizations

Identities are Managed by Entra ID only



Microsoft Entra ID

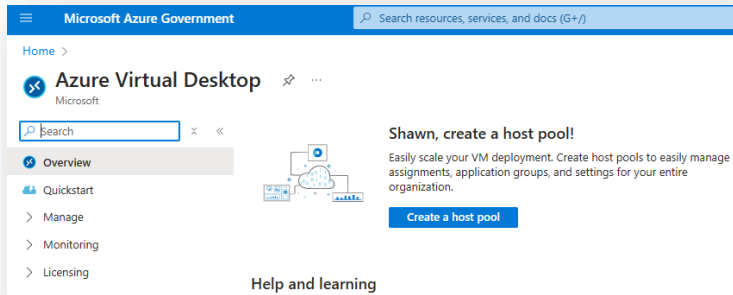
- Join your AVD VMs directly to Microsoft Entra ID
- Connect to VM with basic credentials
- Optionally auto-enroll VMs with Microsoft Endpoint Manager
- Provide access to cloud-only users
- Support for FSLogix profiles currently in preview



Host Pool Deployment

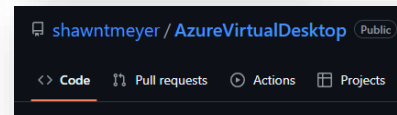
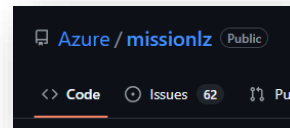


Deployment Options



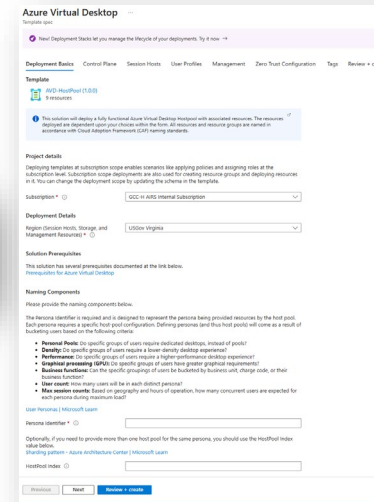
Azure Portal

- Use the Azure Portal to provision a new host pool via a wizard-based process



Custom Options

- Use Microsoft CSA/Fast Track US Gov and Air Gapped Cloud Purpose-Built Solutions for more complete and tailored results



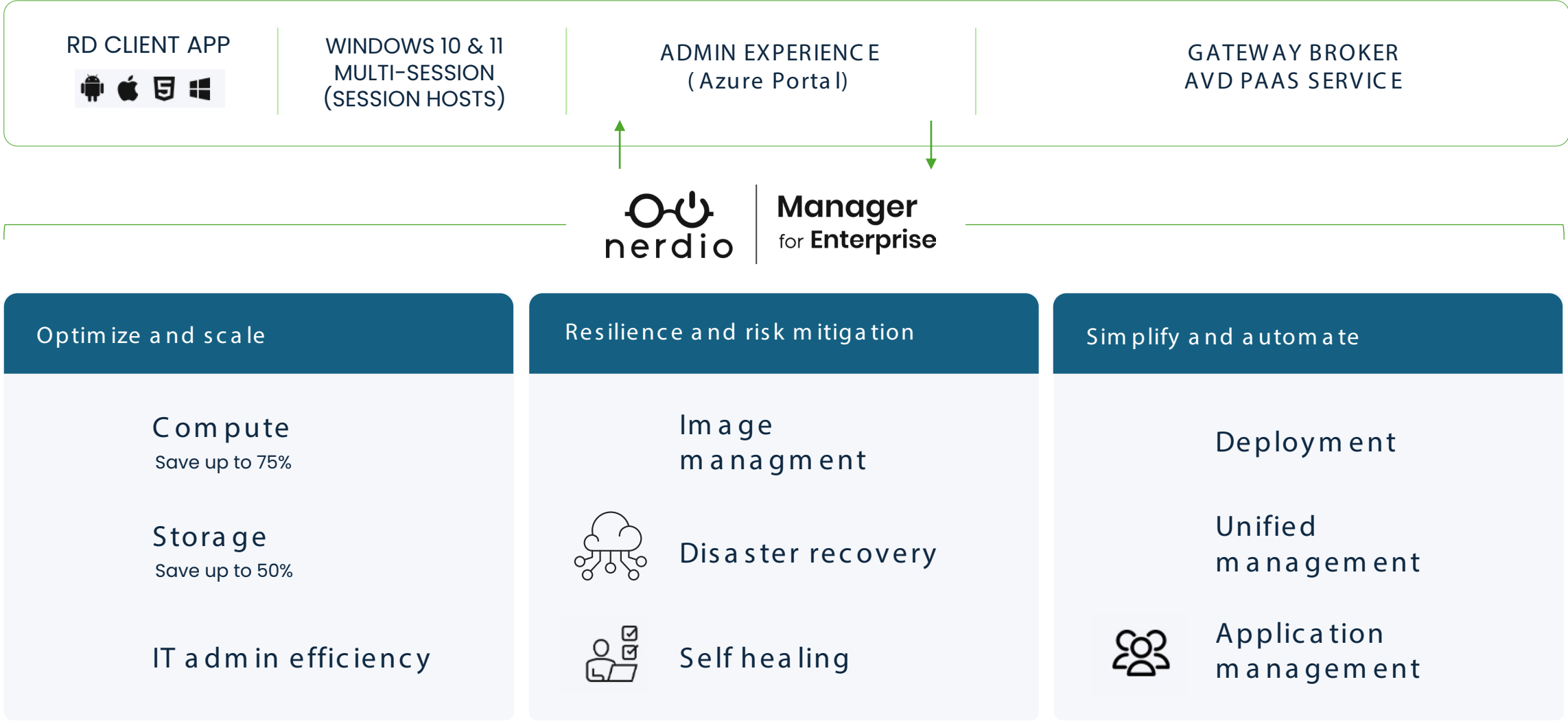
PowerShell / REST API / Partners

- Use your PowerShell Cmdlets or REST API calls to create a host pool



Nerdio Manager unifies the native Microsoft Azure experience for AVD

...



The benefits of Nerdio



Cost insights &
reporting



Image lifecycle
management



Environment
rightsizing



DR & capacity
mitigation



Application
management



Policy
management



HCI stack
support



Modern endpoint
management



End-user self
service

Azure Virtual Desktop gallery images

Azure Virtual Desktop provides a built-in gallery for Windows Enterprise images.

Images with pre-installed M365 Apps

These are images with M365 Apps and Teams optimizations installed by default, ready to go for your users' productivity.

Images with VDI/OS optimizations

These are base Windows Enterprise images with optimizations applied to perform well on virtualized environments.

Note: Use Generation 2 images where possible.

Create a host pool ...

Enable vTPM ⓘ☐

Integrity monitoring ⓘ☒

Image * ⓘ

Select an image ▾

Windows 10 Enterprise multi-session, Version 21H2 - Gen2
Windows 10 Enterprise multi-session, Version 21H2 + Microsoft 365 Apps - Gen2
Windows 10 Enterprise multi-session, Version 22H2 - Gen2
Windows 10 Enterprise multi-session, Version 22H2 + Microsoft 365 Apps - Gen2
Windows 11 Enterprise multi-session, Version 21H2
Windows 11 Enterprise multi-session + Microsoft 365 Apps, Version 21H2
Windows 11 Enterprise multi-session, Version 22H2
Windows 11 Enterprise multi-session + Microsoft 365 Apps, Version 22H2
Windows 11 Enterprise multi-session, Version 23H2 - Gen2
Windows 11 Enterprise multi-session, Version 23H2 + Microsoft 365 Apps - Gen2

Virtual machine size * ⓘ

Number of VMs *

OS disk type * ⓘ

OS disk size * ⓘ

Boot diagnostics ⓘ

Network and security

Review + create

< Previous

Next: Workspace >

Demo 1: Deploy an Azure Virtual Desktop Host Pool





Manage AVD

Profile Management



Understanding profile management

What is a user profile?



A user profile is a collection of settings and information associated with a user. e.g., individual themes, credentials, background, OneDrive sign-in, ...



Typically, every user logging on to an operating system has a locally stored user profile



Profile configuration happens in an organization through GPOs, GPPs, ConfigMgr, CopyProfile, ...

Understanding Profile Management

What does it entail?



It enables a profile to **follow the user** regardless of which device they log on to



Allows **persistent user configurations** on non-persistent machines



Simplifies backup and migration of user profiles

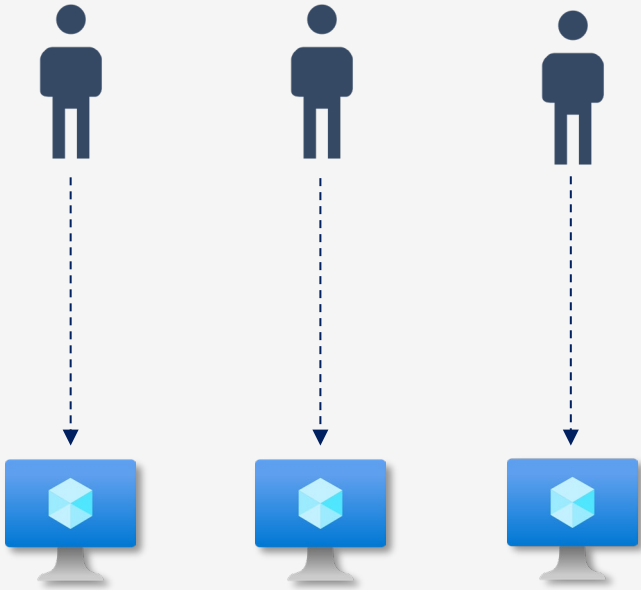


Supports enterprises with **default customizations** to enforce standard configuration



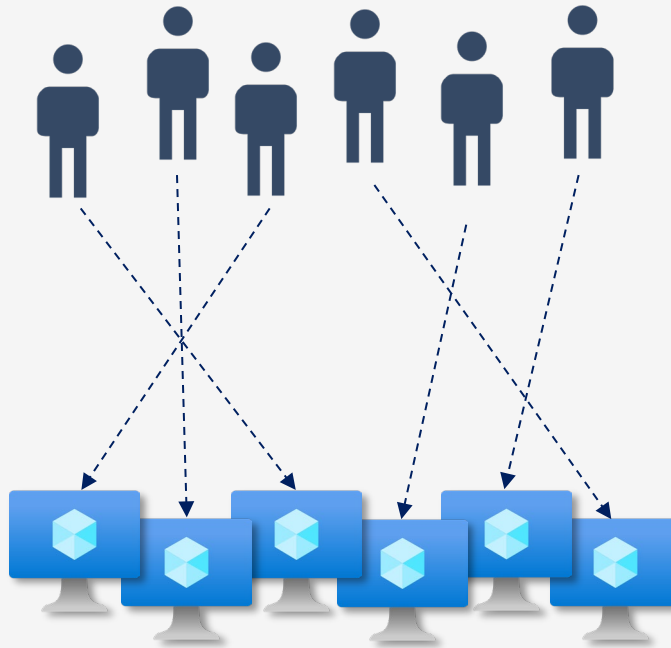
Session-based vs. Machine-based

Personal – Single Session



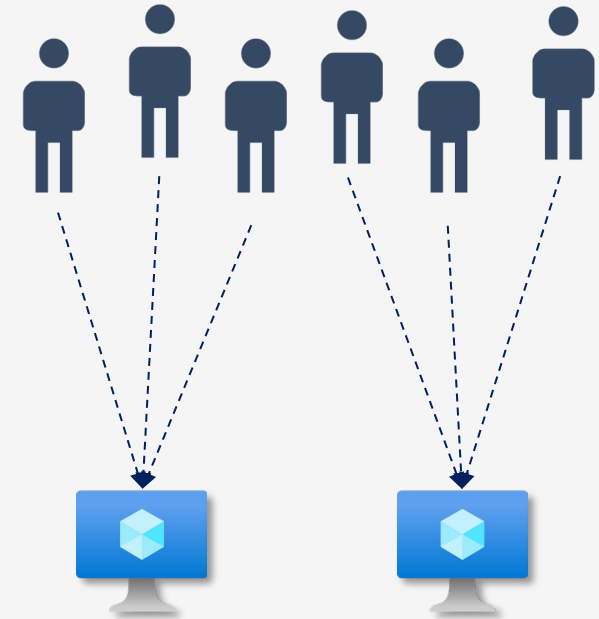
Static
1:1 assignment

Pooled – Single Session



Floating
1:1 usage

Pooled – Multi Session



Multiple users per machine
n:m usage

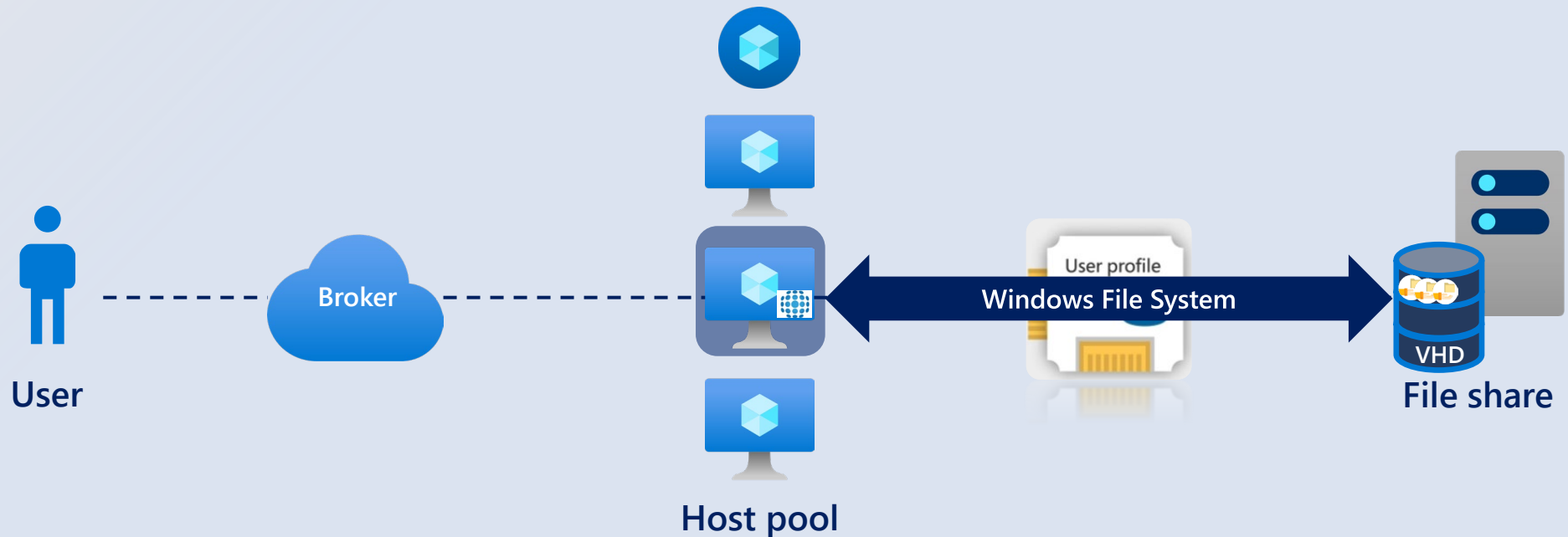
FSLogix Profile Management

Fast login regardless of profile size/efficient IO

Obscures that data is network attached – appears as a local profile

Limited multi-session with read only/diff-disks

Extensive customizations for Office compatibility and performance



Application Management



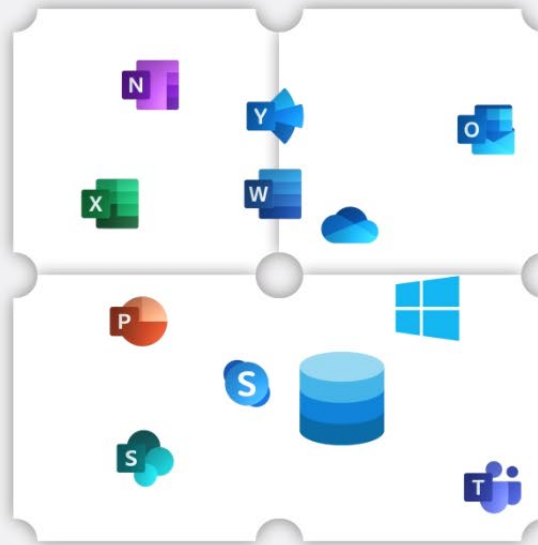
End-user compute environment

3 main components

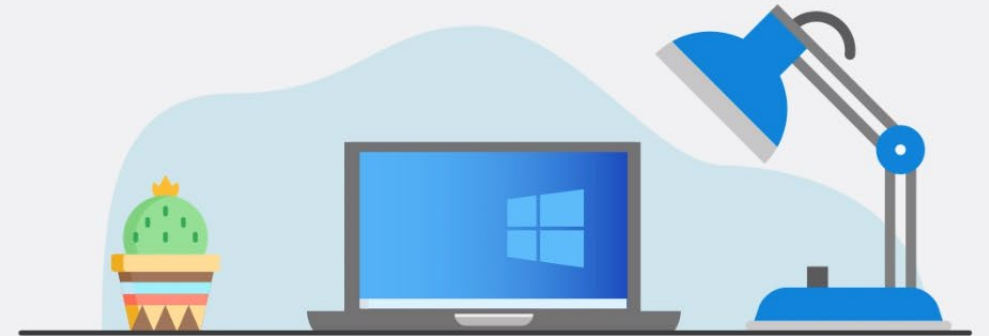
Operating system

Applications

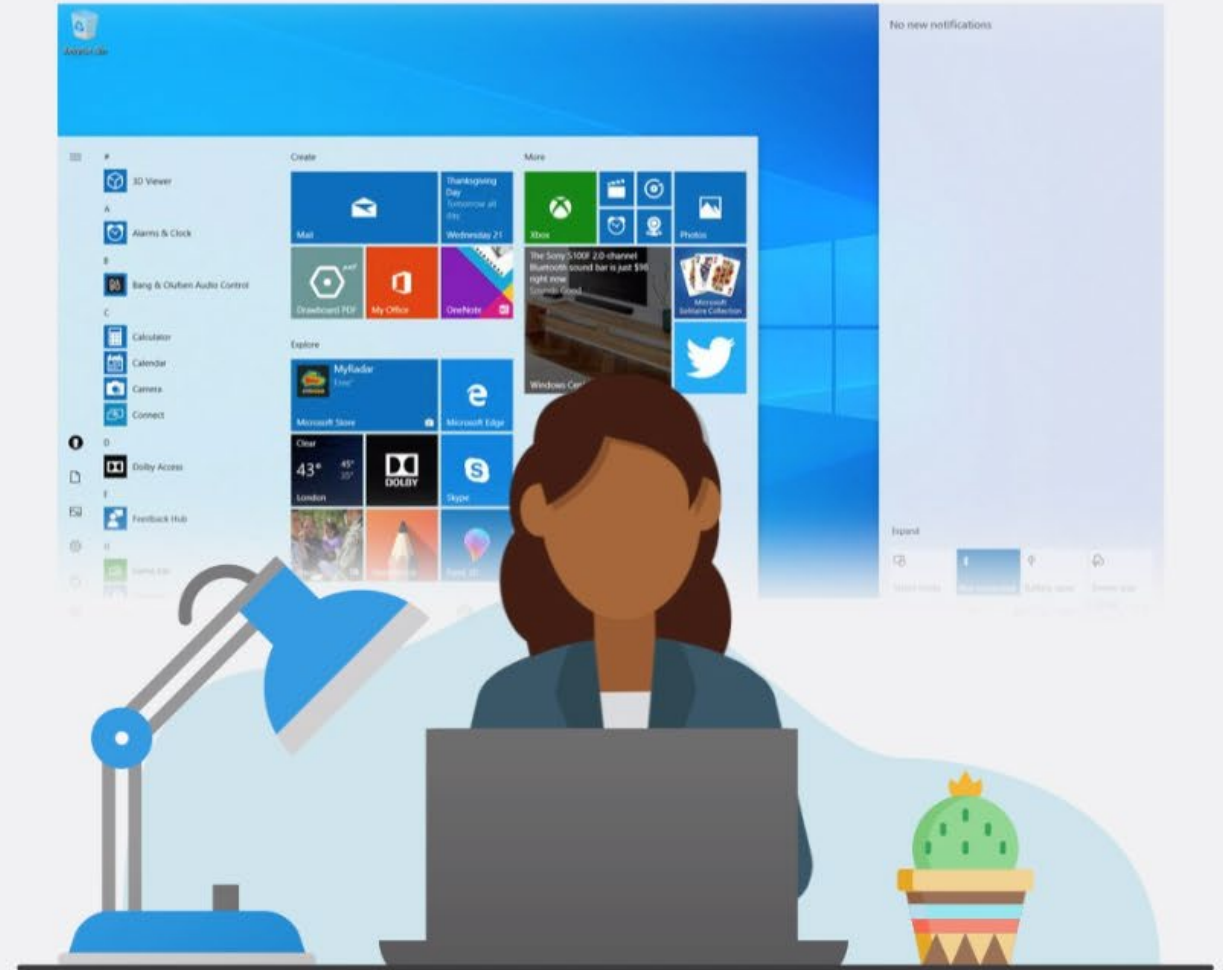
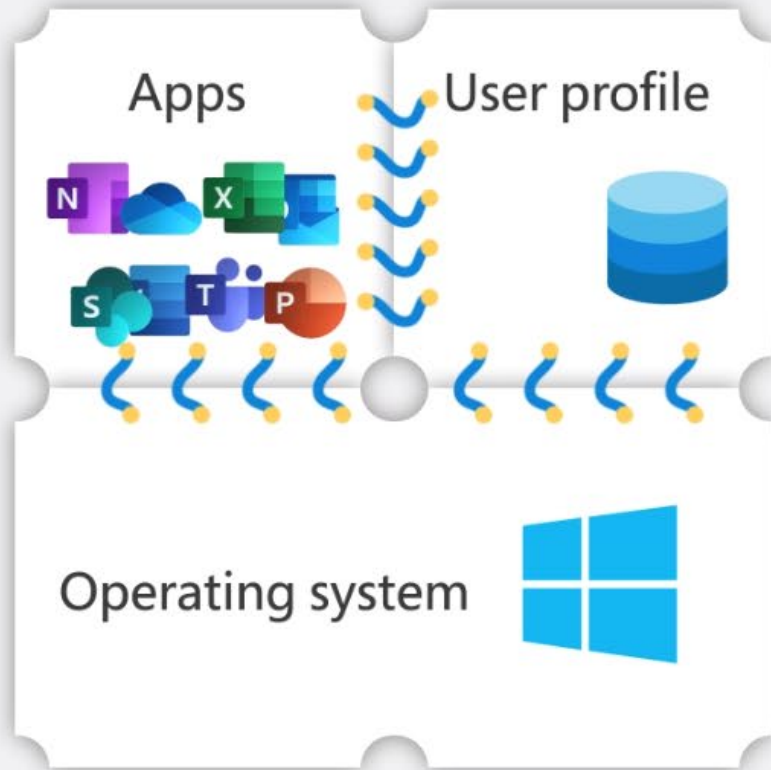
User defined data



Data, applications, and OS



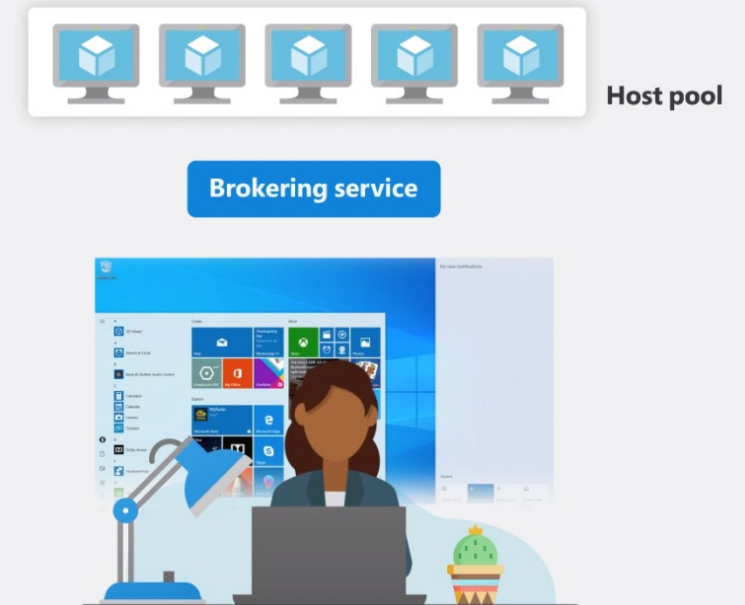
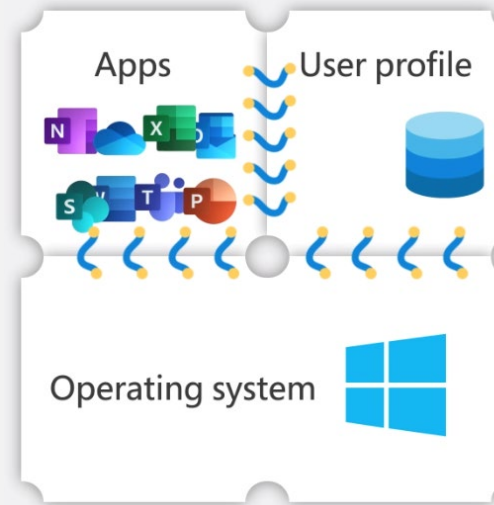
Physical workstation: all components closely coupled



Virtualized compute environments

In an optimized virtualization environment:

A brokering service routes a user to a virtual machine (from a host pool) to a VM with the resources available to host the user's app or desktop workloads.

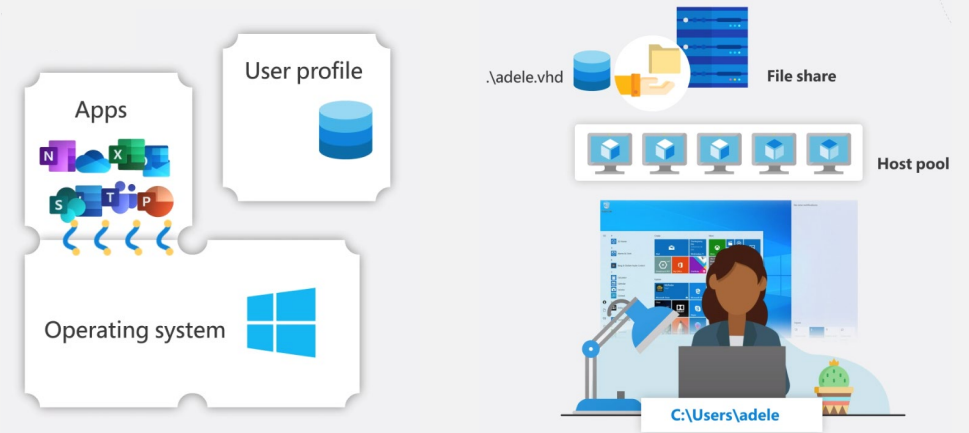


The promise: completely dynamic environments

Containerized user profiles get us half-way there

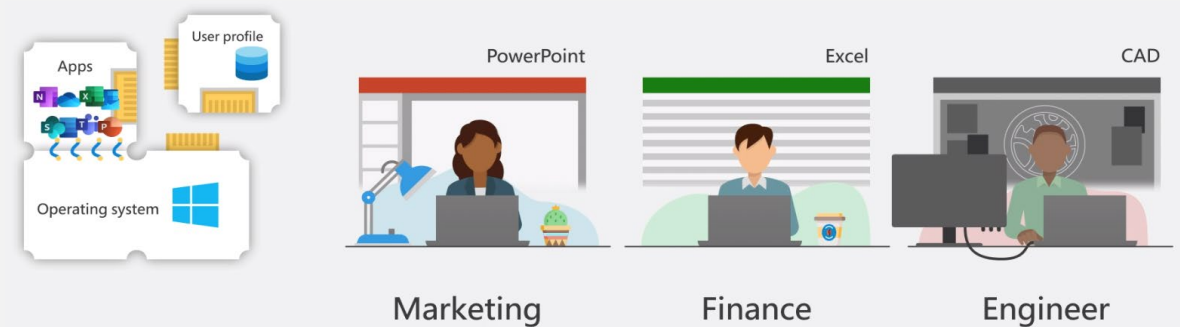
With FSLogix we've separated the user profile layer from the virtual machine

To the user it feels like you're saving and accessing files from a local disk



But - what about the apps?

- In a shared or pooled virtual machine, this is a challenge
- Each user might need a different set of apps
- They can be assigned to a different VM with each logon



App attach

- Native format is MSIX (no re-packaging)
- Minimal performance impact
- MSIX Apps can be stored off the windows disk
- Remotely mount the apps to the VM on-demand
- Apps groups are assigned to users, and they're available instantly on login
- Looks and feels local to the user and to windows

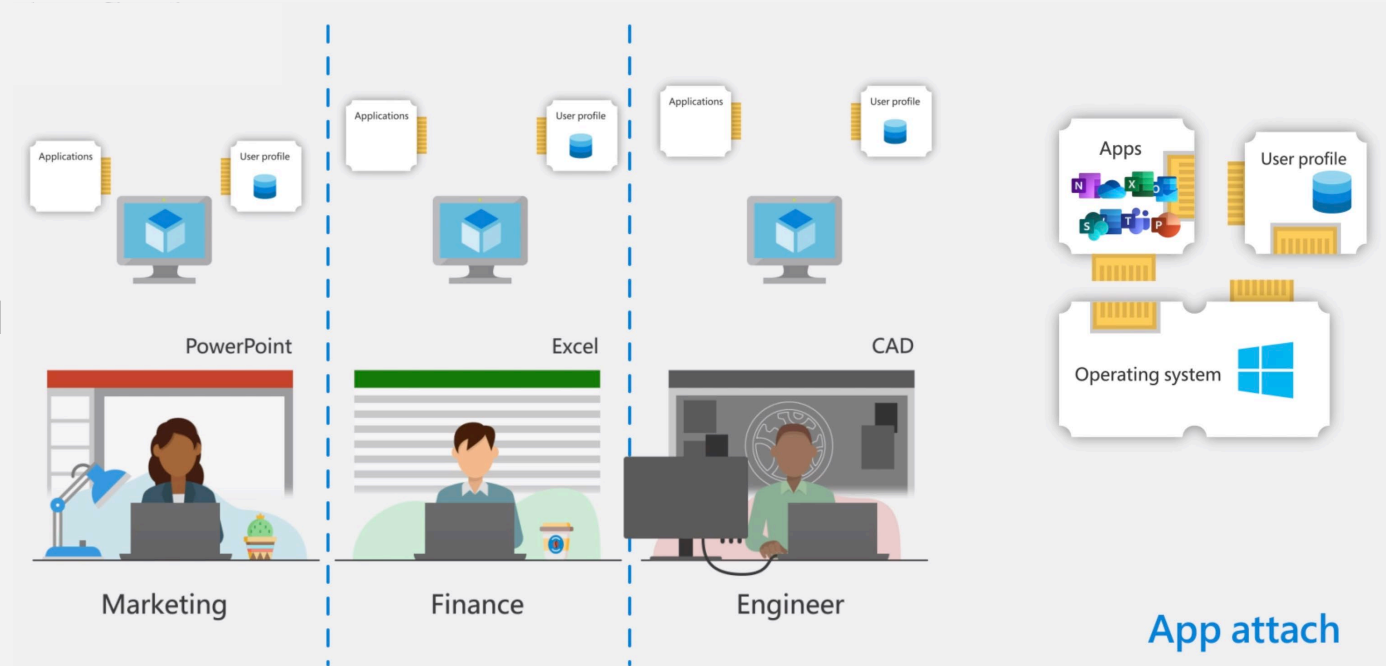


Image Management



The Importance of Image Management

Need to make sure that users have a **secure and consistent environment**

Operating System and applications are **patched and secure**

Application features are **consistent across all VMs**

Customers need to upgrade/add new applications when needed



Image Management

Pooled vs Personal

Personal almost always **managed traditionally**

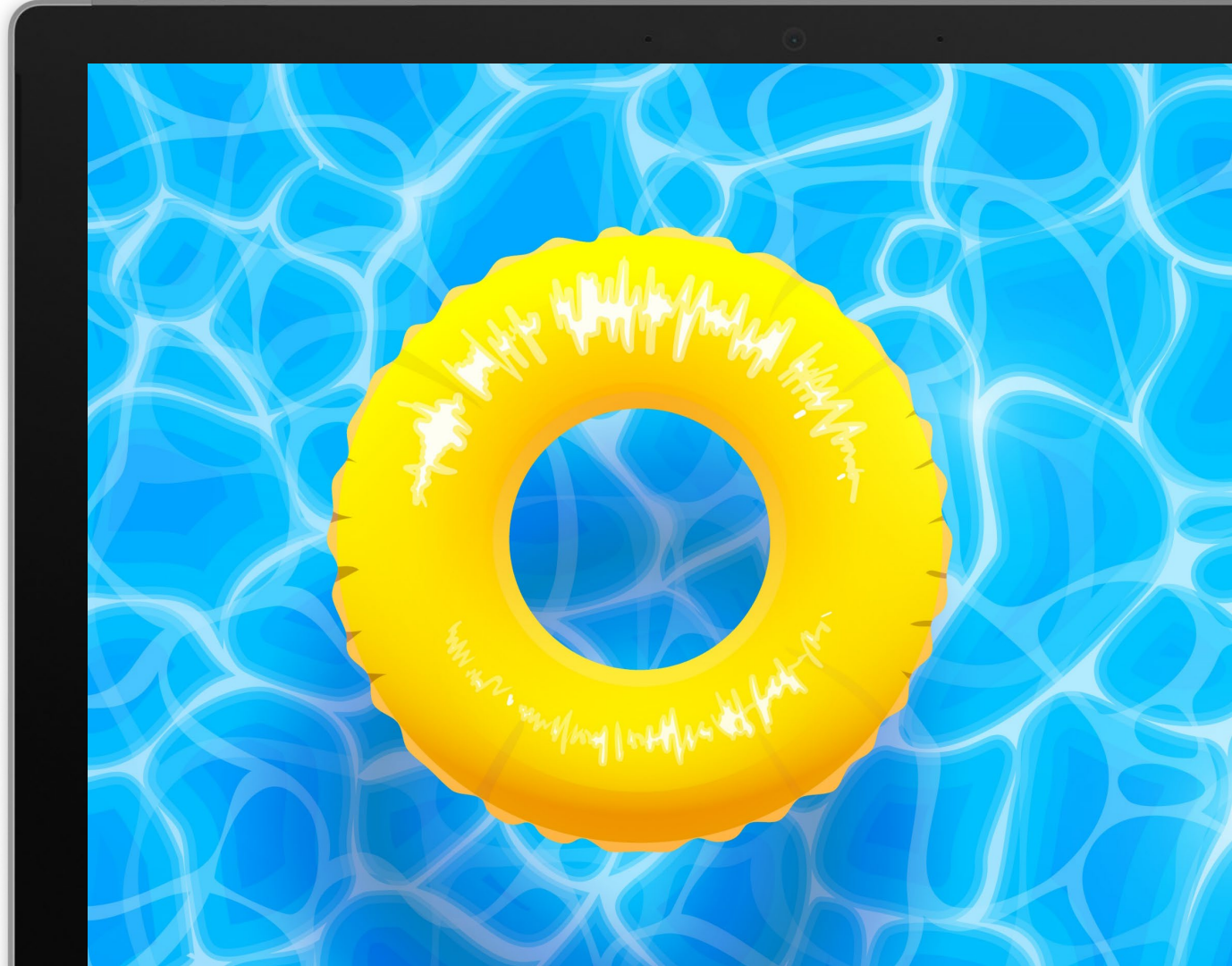
Pooled can gain **large management efficiencies**

Update once in master image and roll out

Roll back anytime

Reset to **known good configuration**

Minimal troubleshooting



Virtual Machine Persistence

Persistent



Azure Virtual Desktop VMs



Administrative changes to VM remain (i.e., software updates, software install, config changes)

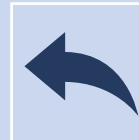


You can manage the Session Hosts or the Image

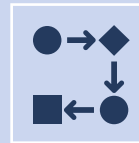
Non-Persistent



VMs built using ephemeral disks (support on roadmap)



VMs revert to image when deallocated and reallocated (OS and software updates do not persist)



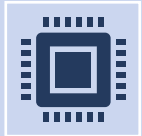
Requires an image management process/strategy

Service Characteristics suggest Management Approach

Pets



A service that does not have an easy scale-out ability, automation, repeatability or transience built into the architecture.



If an individual compute component of the service goes down, you invest the time and resources necessary to get it back to healthy operation.



You treat the individual systems like you would a family pet. You nurse them back to health and perform preventative care.

Cattle



Services running in this service model tend to be distributed across a fleet/farm of servers behind a load balancer.

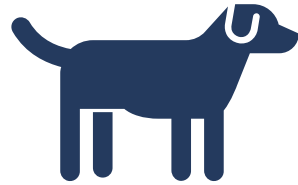


If an instance is unhealthy, you replace it. They are transient, and when they go down, the business doesn't notice.



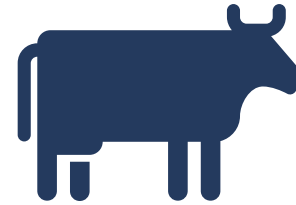
If you need more capacity, you scale out; you add more instances to meet demand.

Pooled Host Pool Session Host Update Options



Pets

Manage the hosts as persistent VMs by patching them and updating software using traditional tools such as Intune.



Cattle

Deploy New host pool based on new image side-by-side

Add session hosts based on the new image to existing host pools

Use automation to 're-image' the session host VMs in place.



Manage the Session Hosts as Pets

Benefits

- Use existing tools and processes
- Easy to understand

Drawbacks

- Doesn't lower TCO
- Increases compliance risk
- Requires coordinating maintenance windows and scaling plans

Side-by-Side Host Pools

Benefits

- Easy to understand and accomplish with automation (BICEP)
- Easy roll back

Drawbacks

- Requires 2 x # of VMs during side-by-side stage.
- Requires feed refresh
- All control plane items must be deployed or updated including workspace, host pool, application group assignments, scaling plans, etc.





Add new VMs to existing host pools

Benefits

- Simple templated deployment of new VMs
- Easy to Understand
- Easy roll back

Drawbacks

- Requires 2 x # of VMs during transition
- Must update tags if scaling in use
- Must update drain mode

Automated Session Host Image Update

Benefits

- Fully automated solution
- Does not require additional VMs

Drawbacks

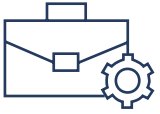
- Requires Function App
- Requires pilot host pool to be updated first with some time for testing
- Roll Back takes longer (redeployment of host pool with old image)



AVD Session Host Replacer [Azure/AVDSessionHostReplacer \(github.com\)](https://github.com/Azure/AVDSessionHostReplacer)

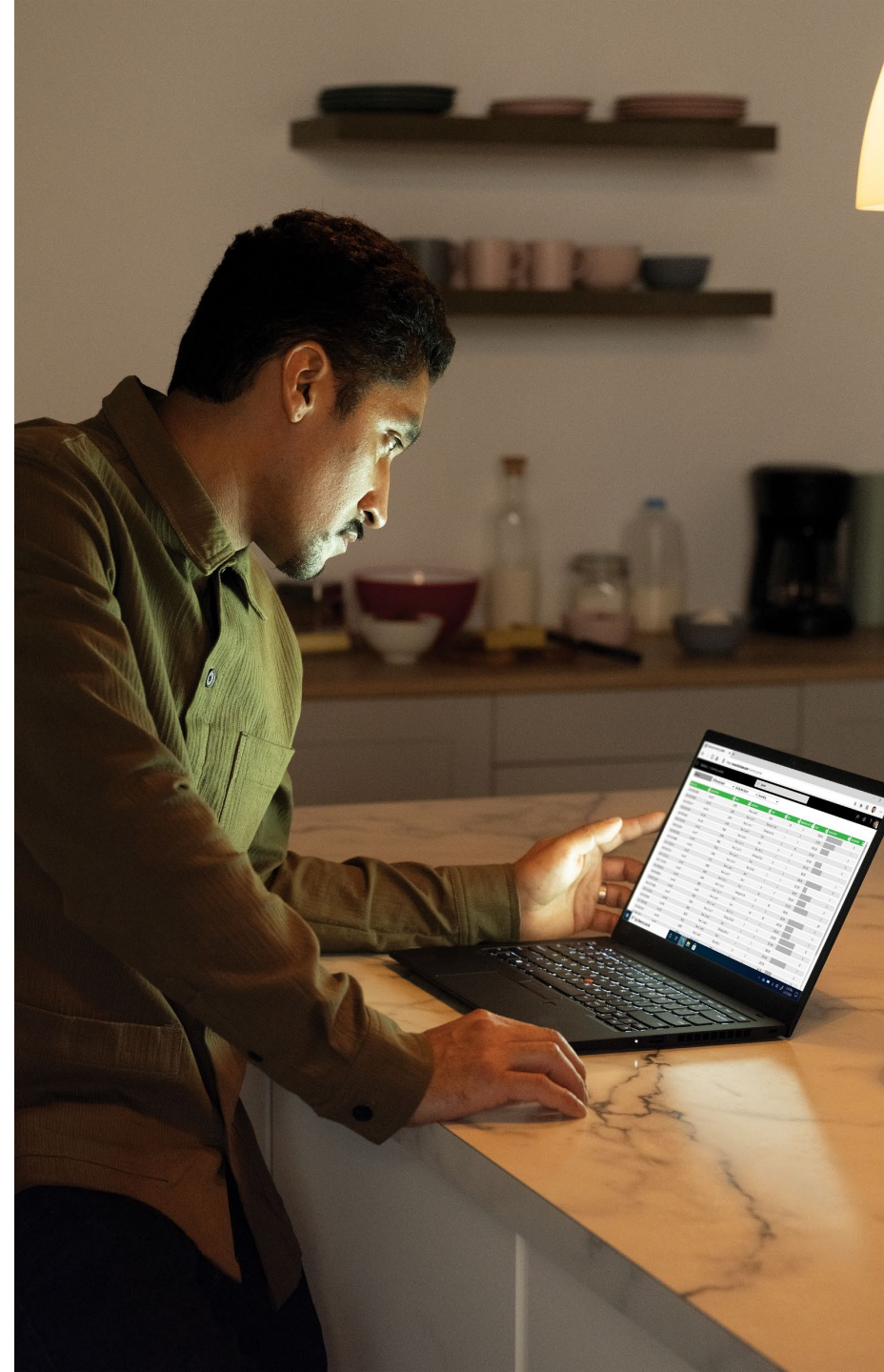
AVD Session Host Update [Session host update - Azure Virtual Desktop | Microsoft Learn](#)

Image Management tooling

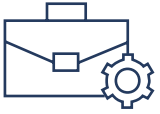


Master images can be created by any already existing process and technologies including

- Azure Image Builder
- Microsoft Endpoint Configuration Manager
- Custom Solutions
- Many 3rd Party Solutions (Nerdio)



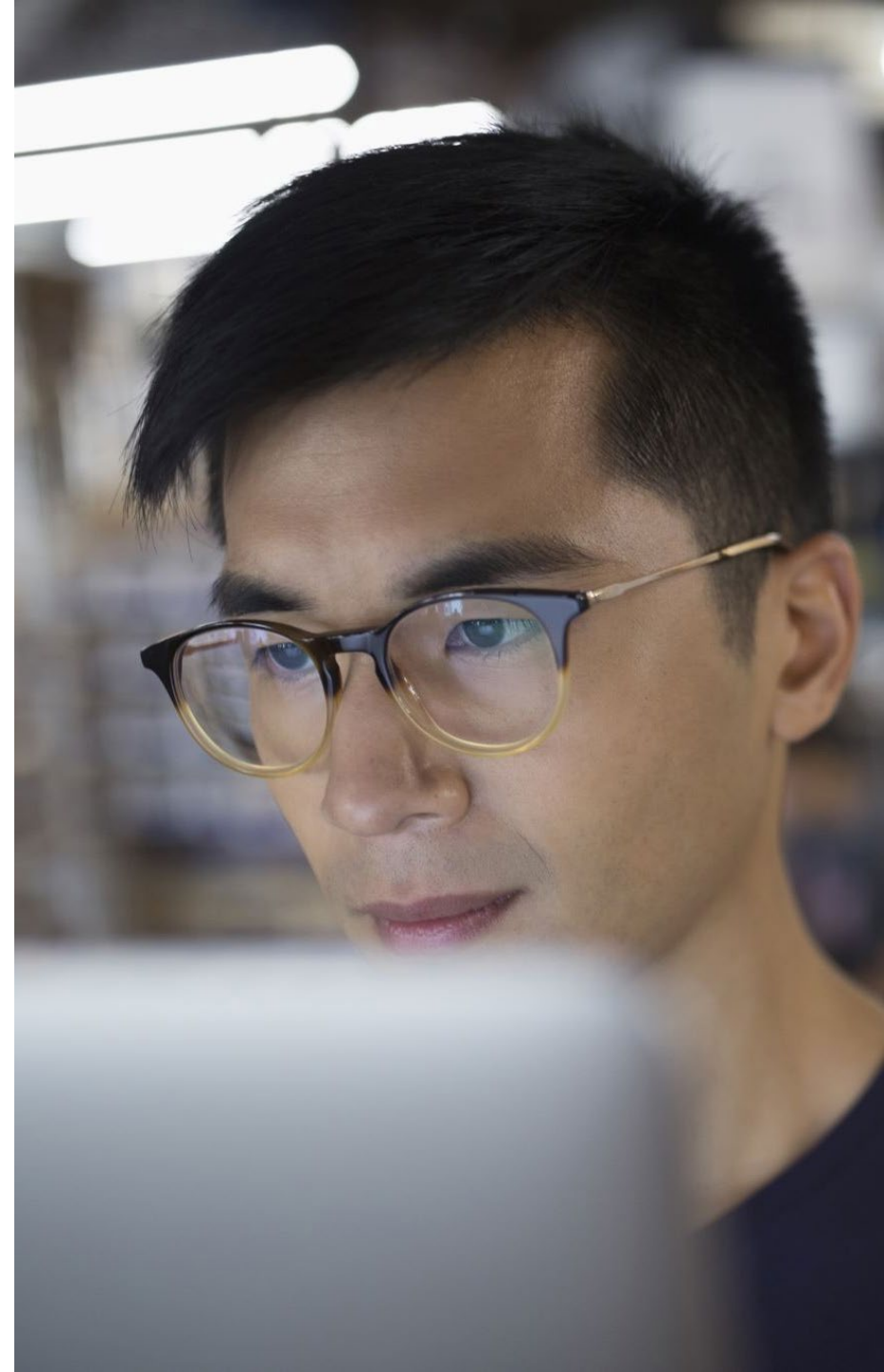
App & Patch Management tooling



Applications and updates can be managed by existing technologies including

- Microsoft Endpoint Configuration Manager/Intune
- Azure Image Builder (break & fix)
- App Masking, App-V and App Attach can be used
- Many 3rd party options

Application deployment strategies exist alongside proper image management, not instead of it.



Azure Image Builder

A new solution for golden image creation



Unification

No multiple image customization pipelines to satisfy different OS's



Simplification

No complexity, less tools, integration into CI/CD pipelines



Management

Handles image lifecycle, no manual copying of blobs between regions



Compliance

Compliant golden corporate images



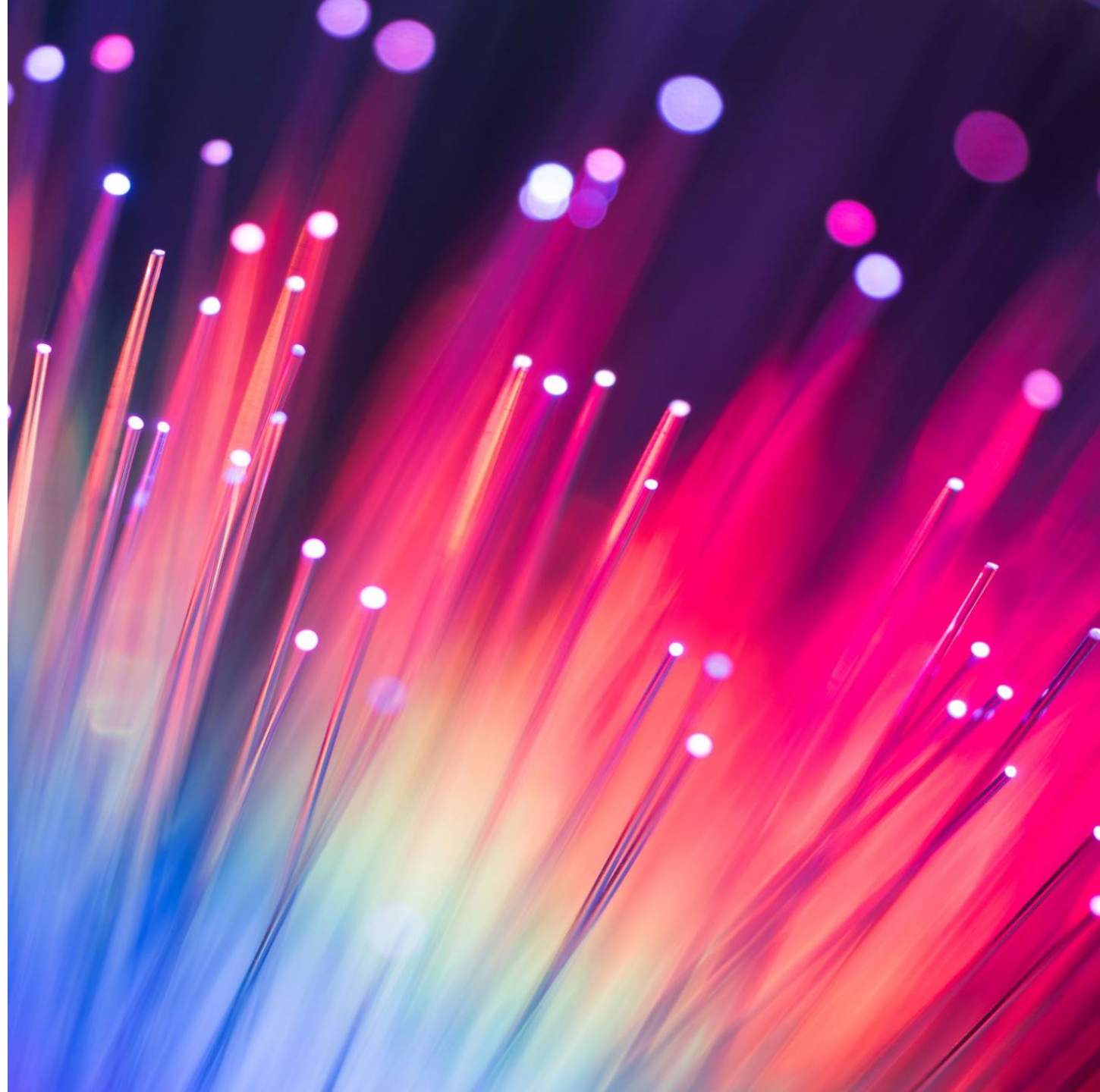
Performance

Extremely fast configuration and deployment



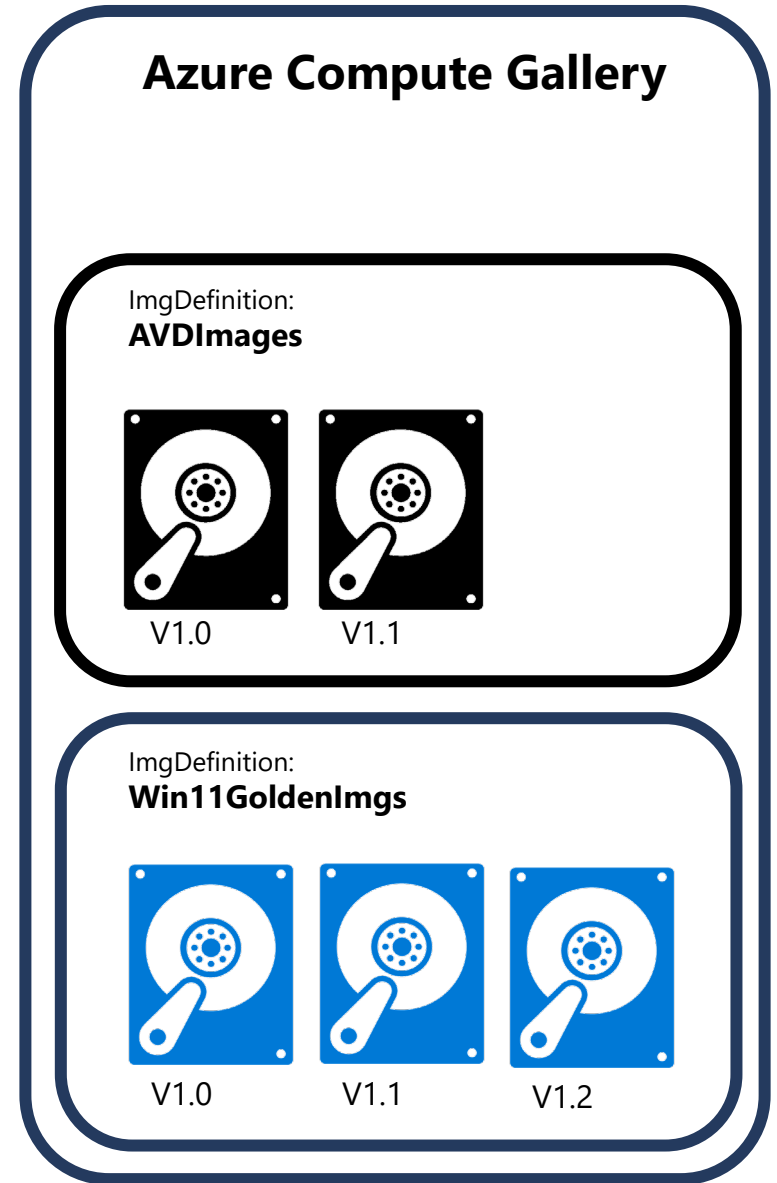
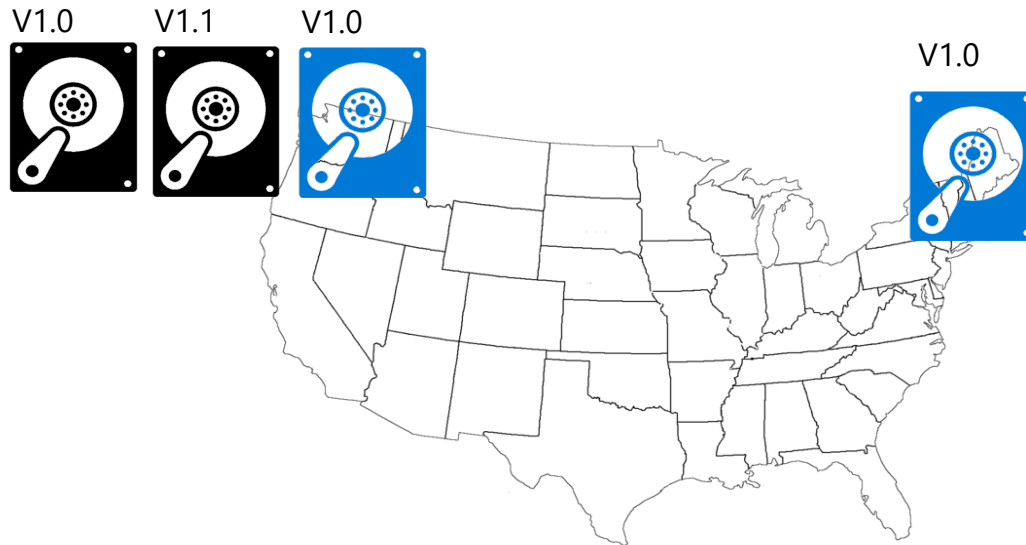
Patching

Update existing images with Security patches



Azure Compute Gallery

- Management of images
- Sharing of images
- Global distribution
- Scaling
- High-availability



Monitoring



Leveraging AVD service diagnostics

End-to-end **user activity tracking**

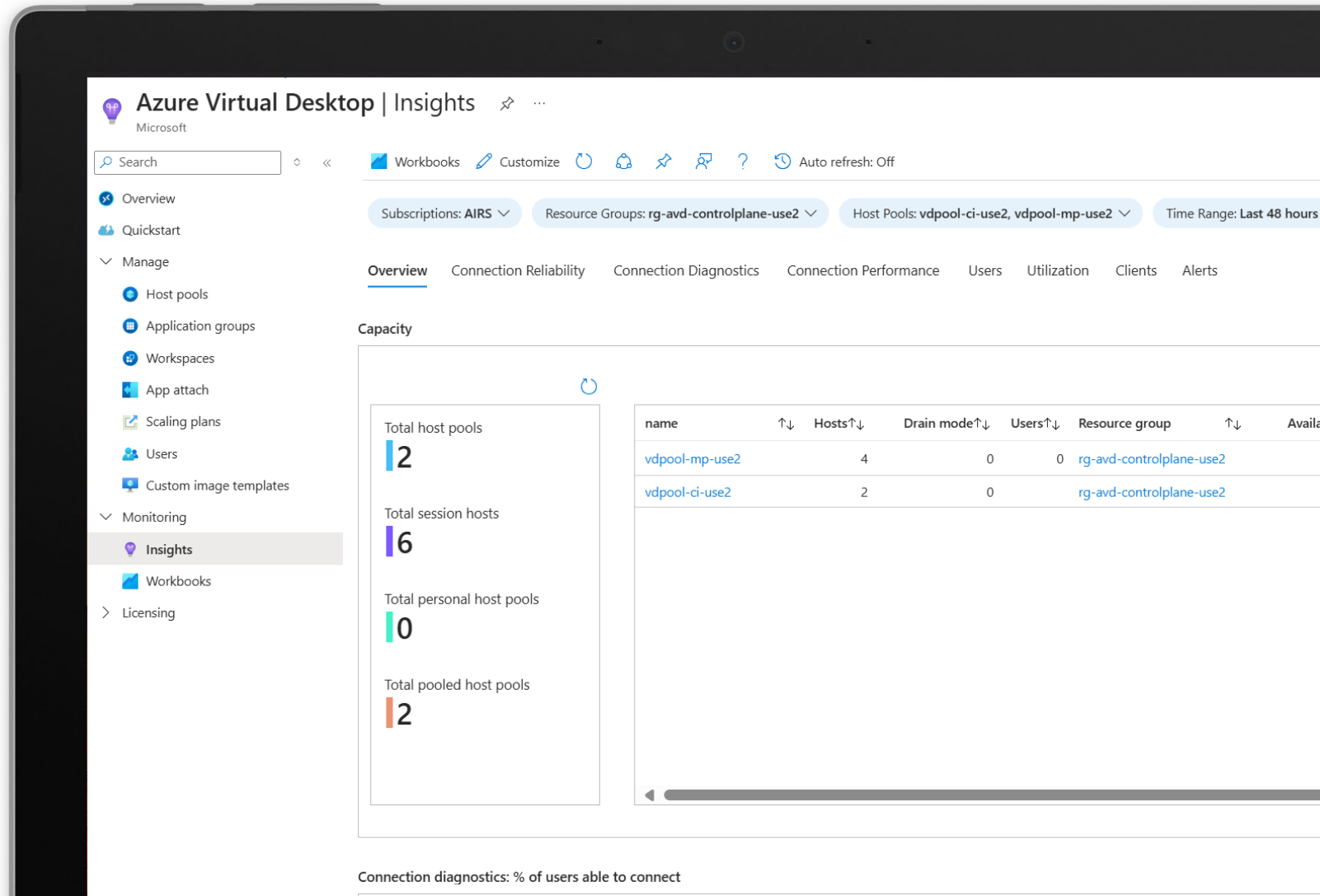
- Administrators managing workspaces
- Users subscribing and connecting to published apps or desktops

Tracks successful **VM registration** with the service

Aggregates and consolidates data into **meaningful information**

Easy access to errors for troubleshooting

Provides **flags** if issue requires escalation to Microsoft Support



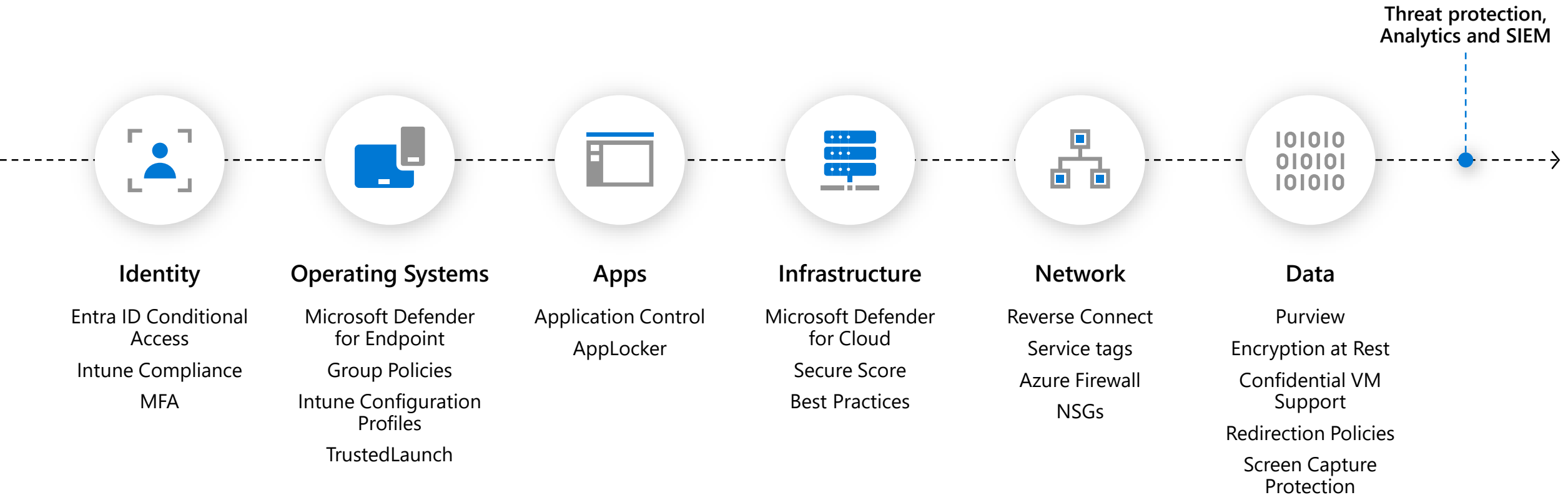
Demo: Azure Virtual Desktop Insights





Secure AVD

End to end security for your virtual desktops



Thank You!

Learn more about Azure Virtual Desktop – Additional sessions Today!

- 10:30AM – 11:00AM
- 12:45PM – 1:30PM
- 1:45PM – 2:30PM
- **3:00PM – 3:45PM**

Introduction to Azure Virtual Desktop
Azure Virtual Desktop: Real World Use Cases
Deploying and Managing AVD
Azure Arc: Optimizing Server Management

- 5:30PM – 6:30PM

Happy Hour sponsored by 

John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com

Shawn Meyer – Principal CSA
Microsoft Federal
shawn.meyer@microsoft.com





FEDERAL AZURE SUMMIT AGENDA

JAN 16 2025

8:30 - 9:00

Registration and Welcome Coffee

9:00 - 9:15

Opening Remarks by [Javier Vasquez](#)

9:15 - 9:45

Keynote: Future of Hybrid Cloud by [Jason Payne](#)

9:45 - 10:15

Adaptive Cloud: Extending Cloud Capabilities to the Edge by [Eric Sanchez](#) / [Joe Fox](#)

10:15 - 10:30

Morning Break

10:30 - 11:00

Introduction to Azure Virtual Desktop (AVD) by [John Kelbley](#) / [Shawn Meyer](#)

11:00 - 11:45

Introduction to Azure VMWare Solution by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

11:45 - 12:45

Lunch Sponsored by



12:45 - 1:30

[Session 1A: Real-World Use Cases of Azure Virtual Desktop](#) by [John Kelbley](#) / [Shawn Meyer](#)

[Session 1B: Real-World Use Cases of Azure Edge Devices](#) by [Rich Shillington](#) / [Jason Benson](#)

1:45 - 2:30

[Session 2A: Deploying and Managing AVD](#) by [Shawn Meyer](#)

[Session 2B: Real-World Use Cases of Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

2:30 - 2:45

Afternoon Break

3:00 - 3:45

[Session 3A: Azure Arc – Optimizing Server Management](#) by [John Kelbley](#)

[Session 3B: Deploying & Managing Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

4:00 - 4:45

Fed Chief Architect Panel: The Future of Cloud and Edge Computing by [Jason Payne](#) / [Chief Architects](#)

4:45 - 5:00

Closing Remarks and Networking by [Javier Vasquez](#)

5:30 - 6:30

Happy Hour Sponsored by



Please visit <https://aka.ms/FedAzSummit/home> for more information

Extend Azure Virtual Desktop with Azure Local



Azure Virtual Desktop for the hybrid cloud

- Traditional virtual desktop infrastructure (VDI) solutions
 - Lack the agility, scale and resilience provided by the cloud
 - Are complex and expensive to operate
- Azure Virtual Desktop is a flexible cloud VDI platform
 - Secure remote access to Windows 10 or 11 virtual desktops
 - Unique multi-session or single session support for cost optimization
 - Less management overhead than traditional VDI
 - Cloud scale, security, agility, and resilience
- But not everything can run in the public cloud
 - Some latency-sensitive workloads
 - Data locality



Introducing

Azure Virtual Desktop for Azure Local

Extending the benefits of cloud-managed VDI to on-premises

Meet unique performance and data locality requirements with on-premises desktop and app virtualization

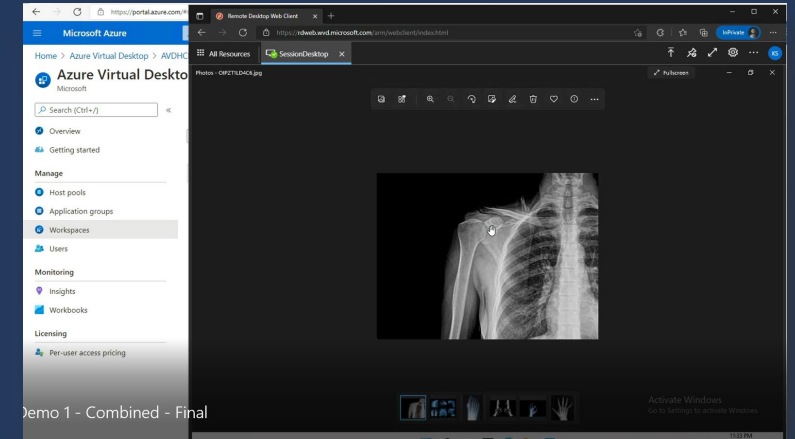
Centralized, cloud-based management for virtual desktops and applications

The only VDI solution with full Windows 10 or 11 multi-session support for cost optimization

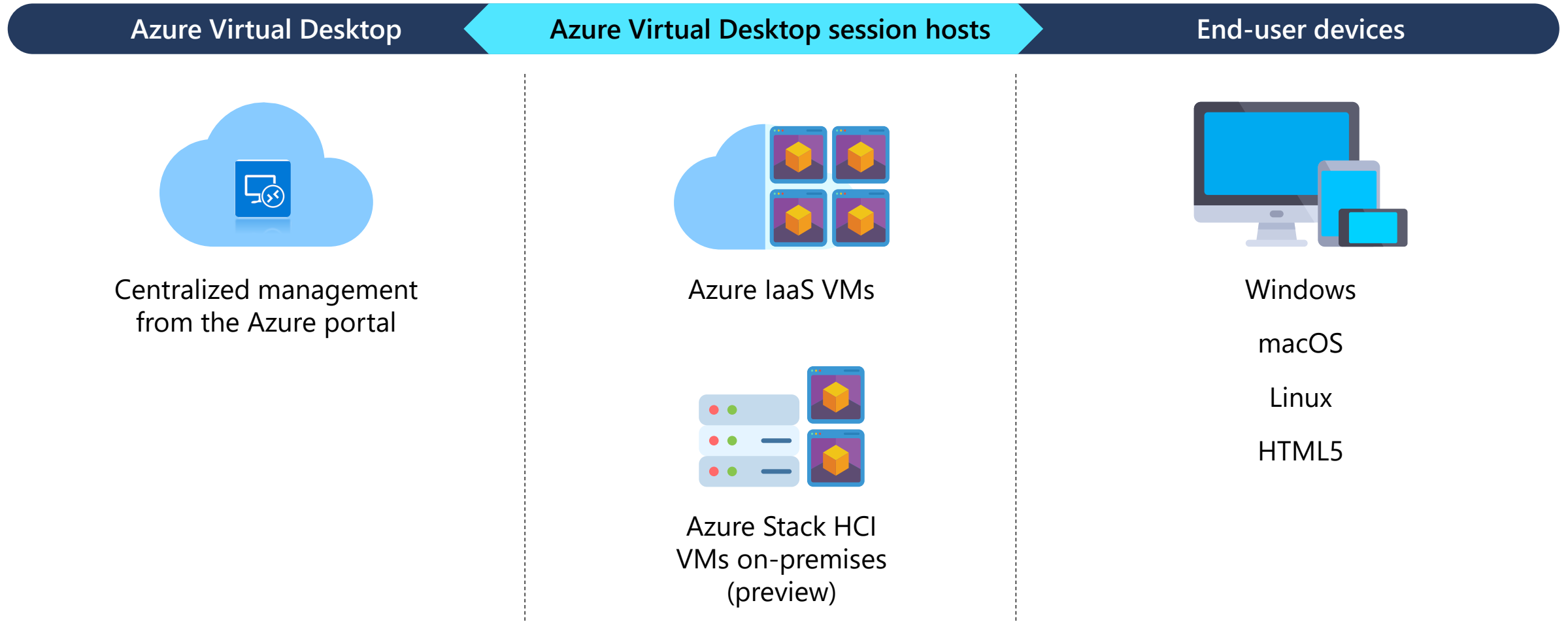
Windows server 2022 and 2019 Support

Seamless integration with Microsoft 365 and Teams

End-user access from virtually anywhere



Centralized management for Azure Virtual Desktop



Benefits of Azure Virtual Desktop for Azure Local

Cloud-based VDI



- Simplify your VDI deployment—no need to manage brokers, gateways, or underlying servers and storage

Windows 11 multi-session



- Get Windows 10 & Windows 11 multi-session or single session support
- Achieve high utilization & lower operation costs

Performance



- Enjoy optimized Microsoft 365 / Teams experience
- Use RDP ShortPath for low latency user access
- Run graphic-intensive workloads with GPU support

Full control



- Satisfy data locality requirements with efficient, performant on-premises storage and DR

Scale across cloud and on-premises



- Manage and scale deployments across both Azure and Azure Stack HCI through a single management experience

Optimize for cost



- Use existing eligible Windows licenses
- Save with Windows 10/11 multi-session support

How to try Azure Virtual Desktop for Azure Local



Buy a validated hardware solution or re-use existing hardware if it matches a solution in the Azure Stack HCI Catalog

[Browse solutions](#)



Install Azure Stack HCI software with a free 60-day trial

[Download software](#)



Visit the [documentation page](#) to install Azure Virtual Desktop

No charge during preview

Please note:

1. An Azure subscription is required to complete the full set-up on your server hardware
2. Integrated systems (and some validated nodes) come with Azure Stack HCI software pre-installed

Security controls and best practices

Identity



Require Multifactor Authentication

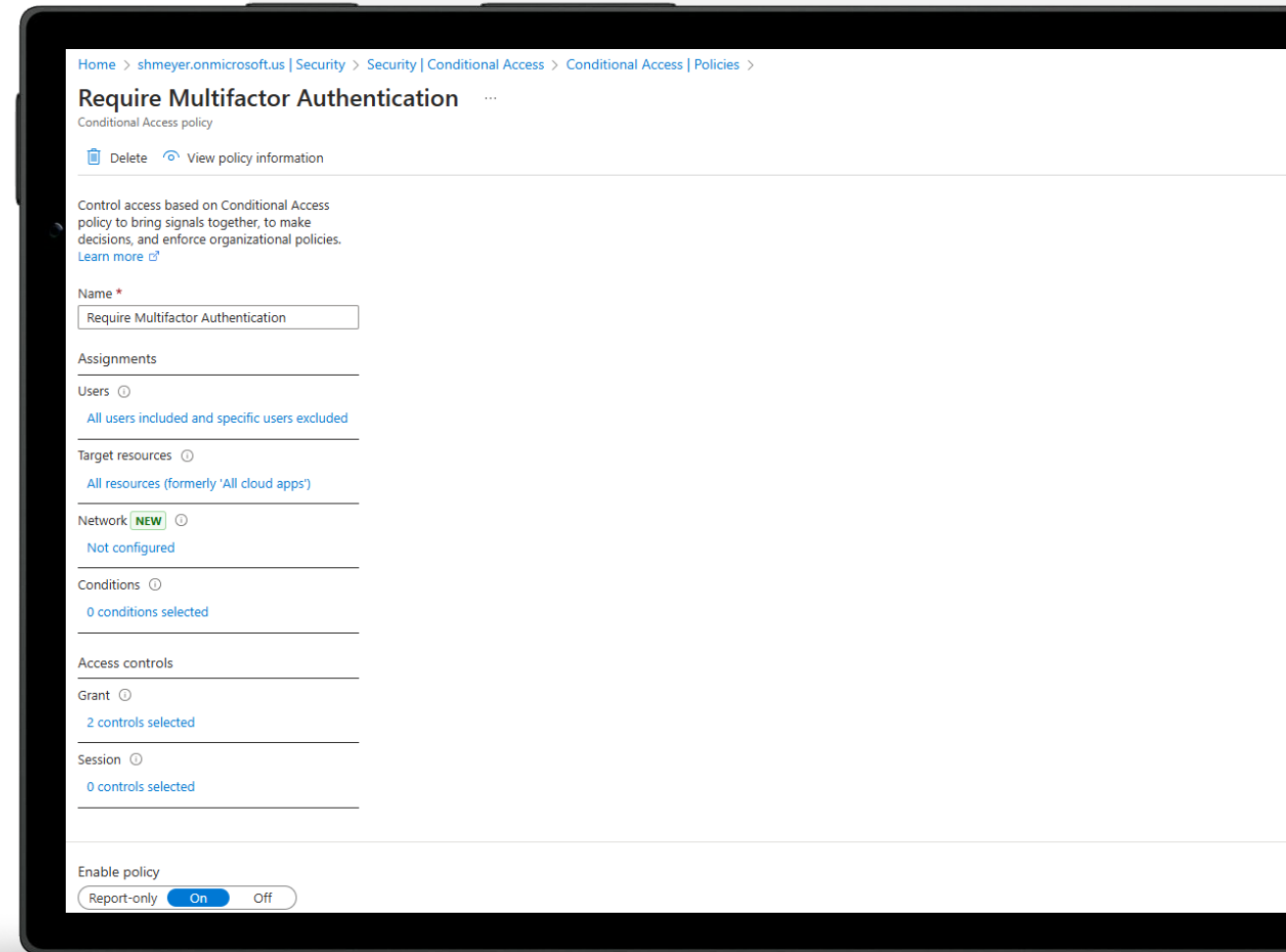
Activate Azure MFA for Microsoft Entra accounts

Enable Conditional Access

Configure a Conditional Access Policy and target AVD

Intune support

<https://aka.ms/AVDSecurityGuide>



Security controls and best practices

Operating Systems



Patch software vulnerabilities

Update live host/redeploy using latest gallery image

Control device redirection

Look into disabling Smart card, Port, Drives, Camera redirection

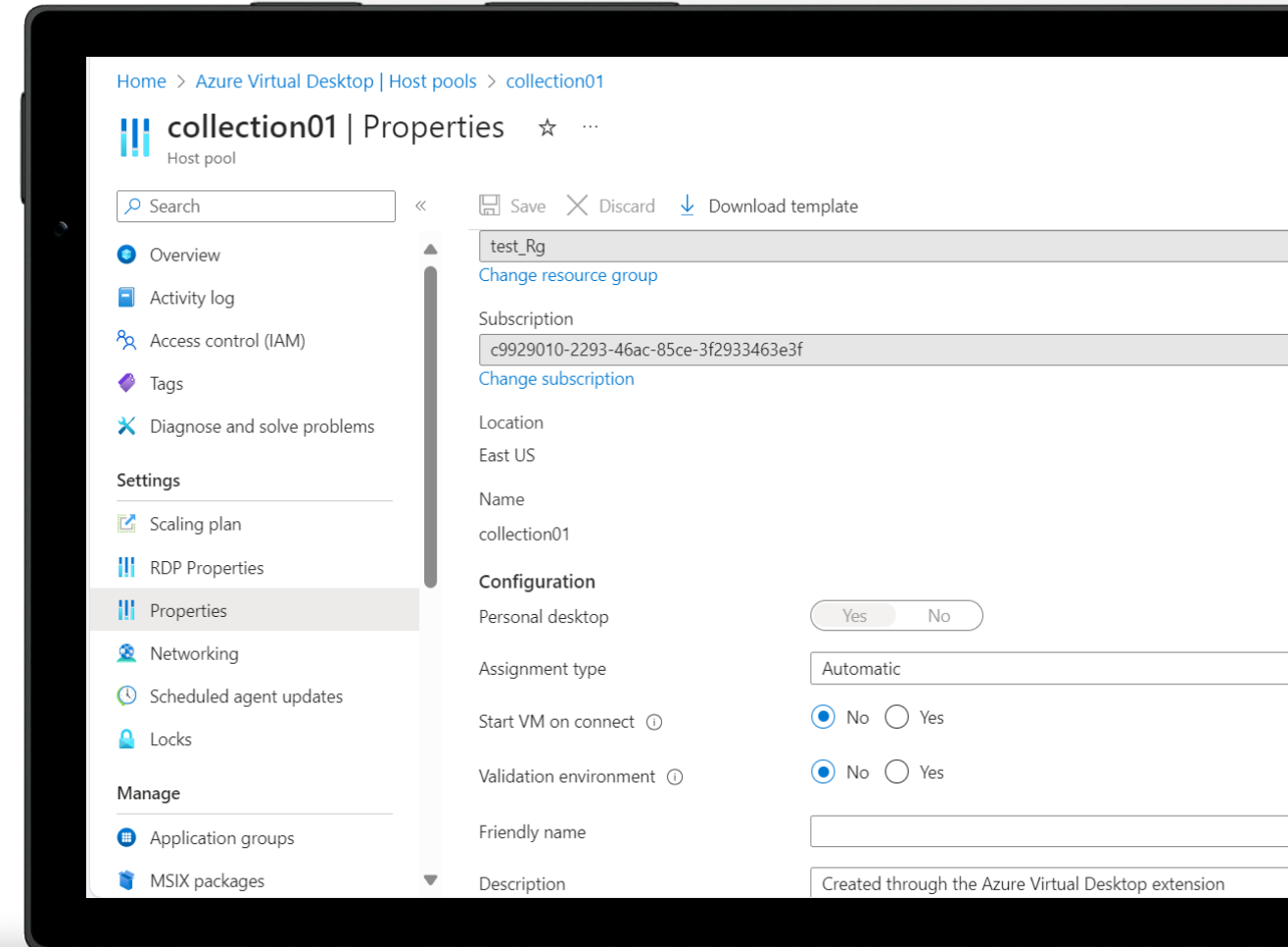
Windows security baseline

Apply Windows 10/11 Security Baselines

Define Group Policies

Set Time Limit For Active But Idle Remote Desktop Services Sessions

Set time limit for disconnected sessions



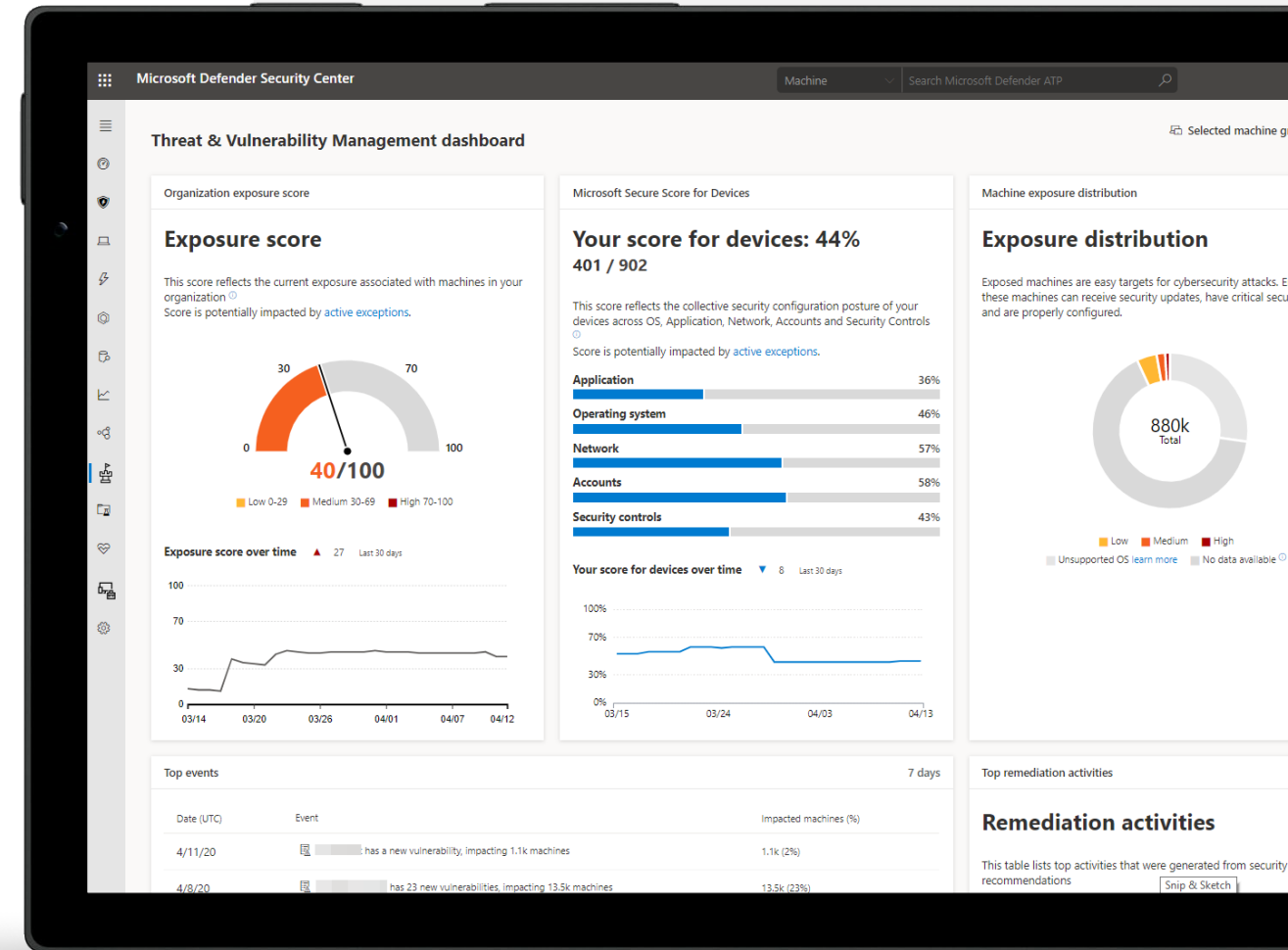
Security controls and best practices

Operating Systems



Microsoft Defender for Endpoint

- Next-generation Antivirus
- Endpoint Detection and Response
- Network Protection, Web Content Filtering, Attack Surface Reduction
- Threat Vulnerability Management



Security controls and best practices

Apps



AppLocker

Control what applications users can run

Application Control

Control what drivers and applications can run

<https://aka.ms/wvdappssec>

The screenshot shows a web page with a left-hand navigation pane and a main content area. The navigation pane includes a search bar labeled 'Filter by title' and a list of links: 'Application Control for Windows', 'About application control for Windows' (expanded), 'About application control for Windows', 'WDAC and AppLocker Overview' (highlighted), 'WDAC and AppLocker Feature Availability', 'Virtualization-based protection of code integrity', 'WDAC design guide', 'WDAC deployment guide', 'WDAC operational guide', 'WDAC AppId Tagging guide', and 'AppLocker'. The main content area has the title 'Choose when to use WDAC or AppLocker' and contains the following text:

Generally, customers who are able to implement application control using WDAC, rather than AppLocker, should do so. WDAC is undergoing continual improvements, and is getting added support from Microsoft management platforms. Although AppLocker continues to receive security fixes, it isn't getting new feature improvements.

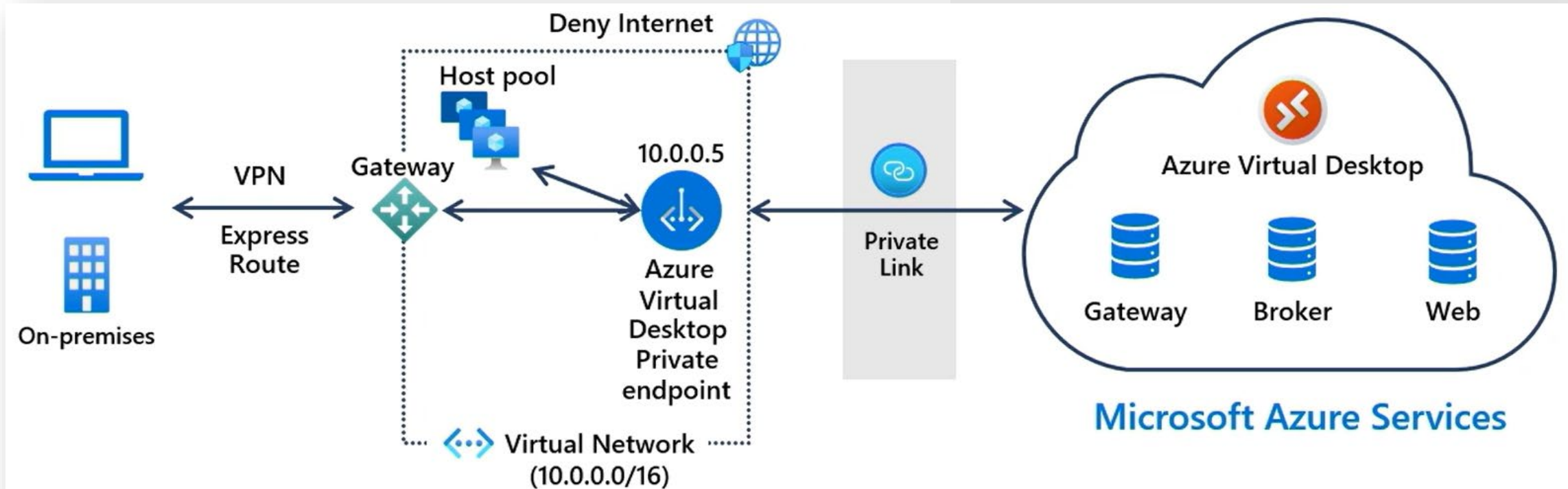
However, in some cases, AppLocker might be the more appropriate technology for your organization. AppLocker is best when:

- You have a mixed Windows operating system (OS) environment and need to apply the same policy controls to Windows 10 and earlier versions of the OS.
- You need to apply different policies for different users or groups on shared computers.
- You don't want to enforce application control on application files such as DLLs or drivers.

AppLocker can also be deployed as a complement to WDAC to add user or group-specific rules for shared device scenarios, where it's important to prevent some users from running specific apps. As a best practice, you should enforce WDAC at the most restrictive level possible for your organization, and then you can use AppLocker to further fine-tune the restrictions.

Security controls and best practices

Network



What is Private Link for AVD?

Private Link for Azure Virtual Desktop allows users to establish secure connections to remote resources by using private endpoints.

Understanding benefits

Private Link gives customers added security, allowing access to remote resources via private endpoints within the protected Microsoft Backbone network.

Getting started

Follow these steps to [configure Private Link for Azure Virtual Desktop](#)

Security controls and best practices

Network



Reverse Connect

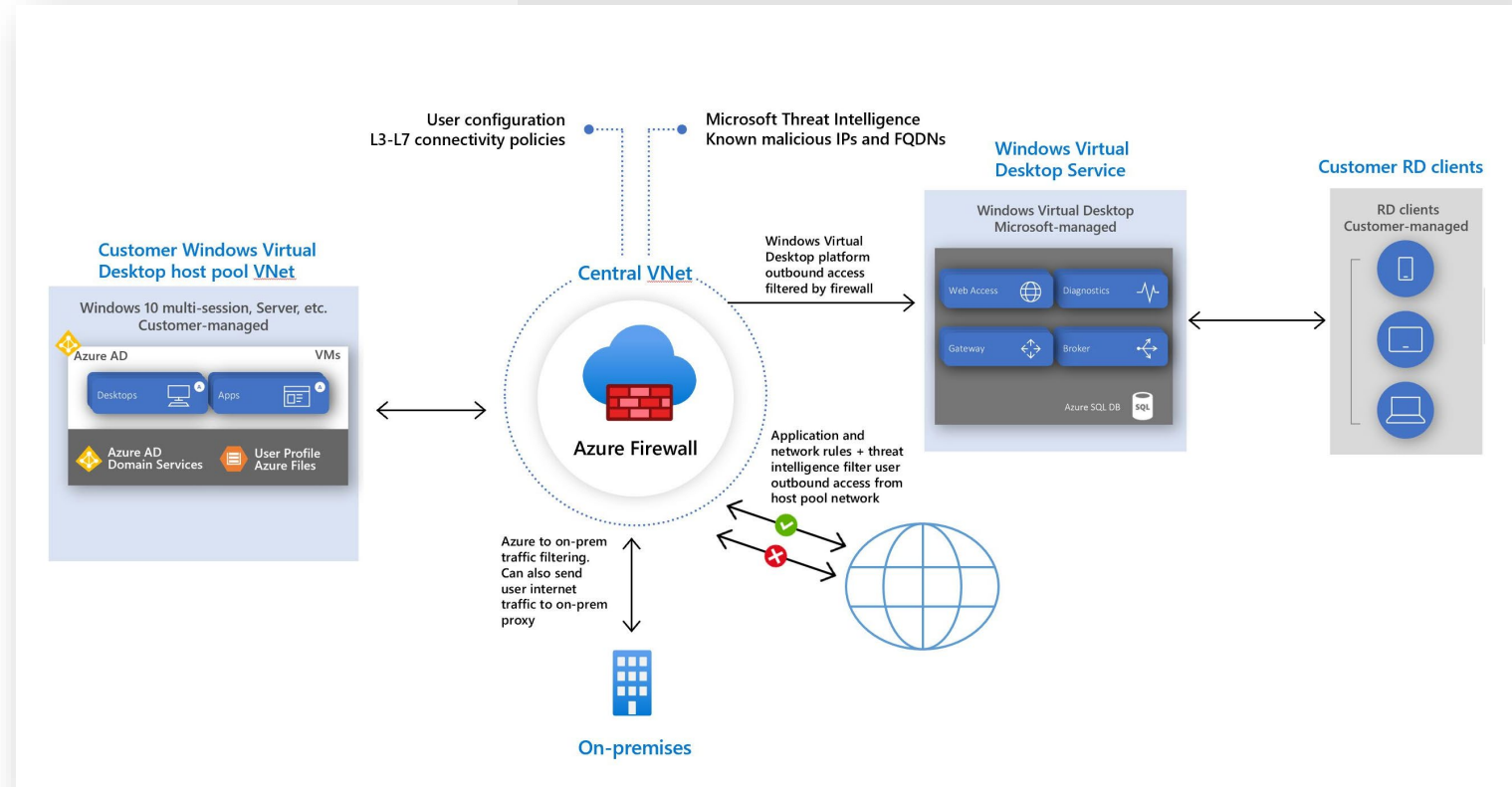
Disable all inbound traffic

NSG Firewall service tags

Limit network-level traffic Azure Virtual Desktop traffic with service tags

Azure Firewall

Consider Azure Firewall for application-level protection with Azure Virtual Desktop FQDN tag



<https://aka.ms/wvdfirewall>

Security controls and best practices

Infrastructure



Enable Azure Defender and Microsoft Defender for Cloud (formerly Azure Security Center)

Provides threat and vulnerability management assessments

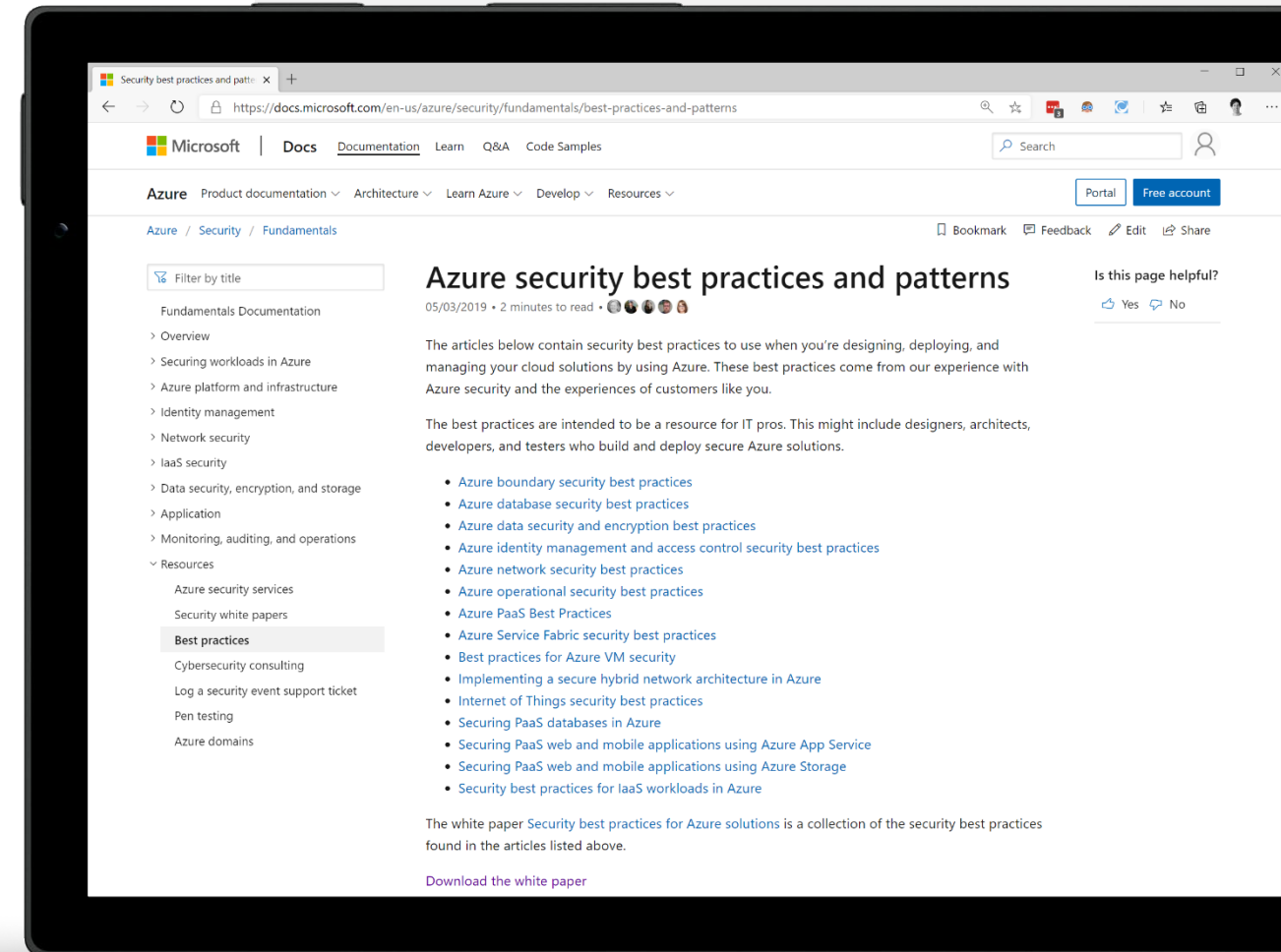
Operationalize your Secure Score

Secure Score provides recommendations and best practice advice for increasing your security posture

Follow Azure best practices

Secure surrounding infrastructure with documented best practices

<https://aka.ms/azuresecurebp>



Security controls and best practices

Data

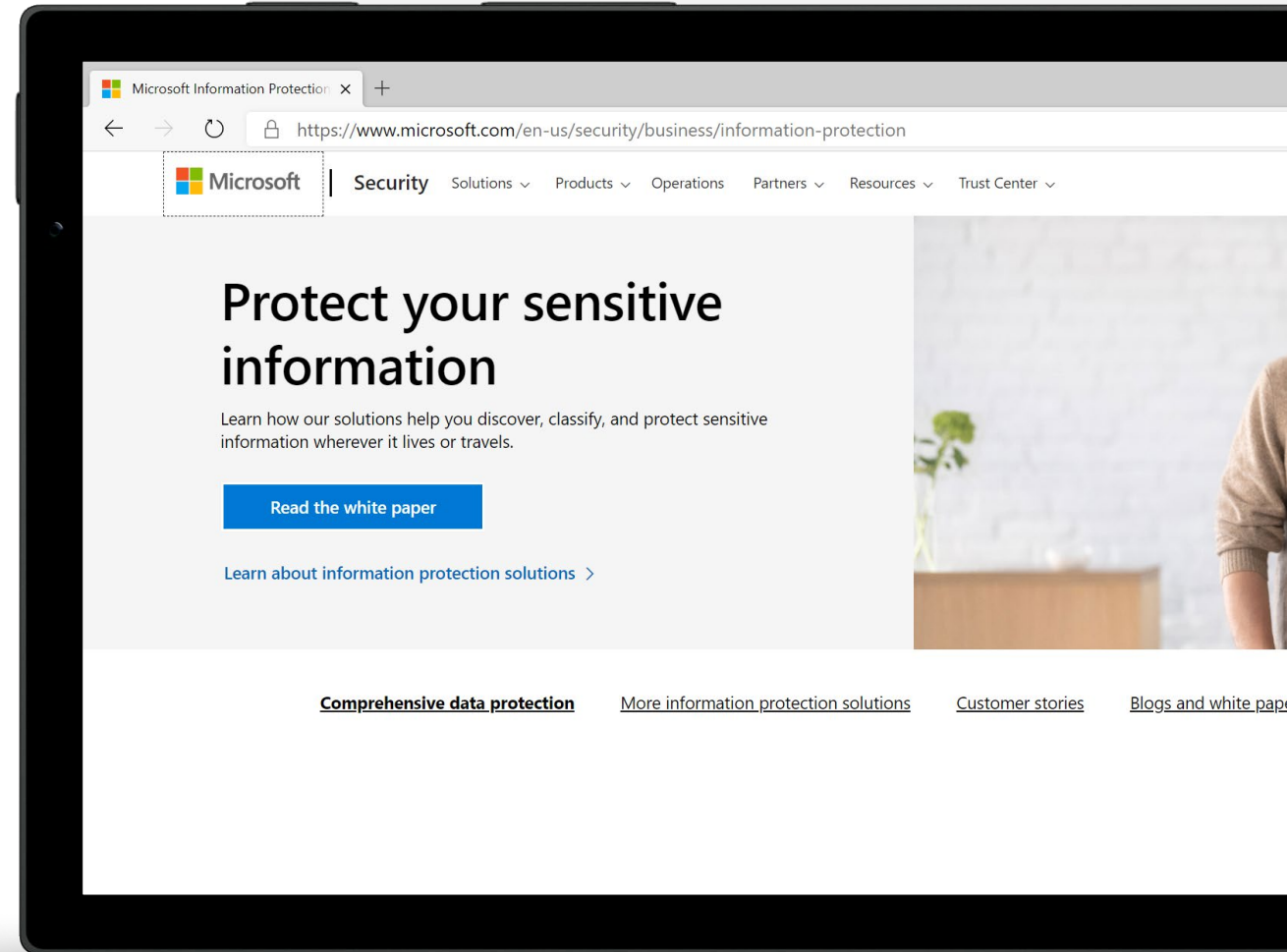
101010
010101
101010

Microsoft Purview Information Protection

Discover, classify, and protect sensitive information wherever it lives or travels

Encryption at Host

Helps protect and safeguard your data



Security controls and best practices

Data



Clipboard transfer direction

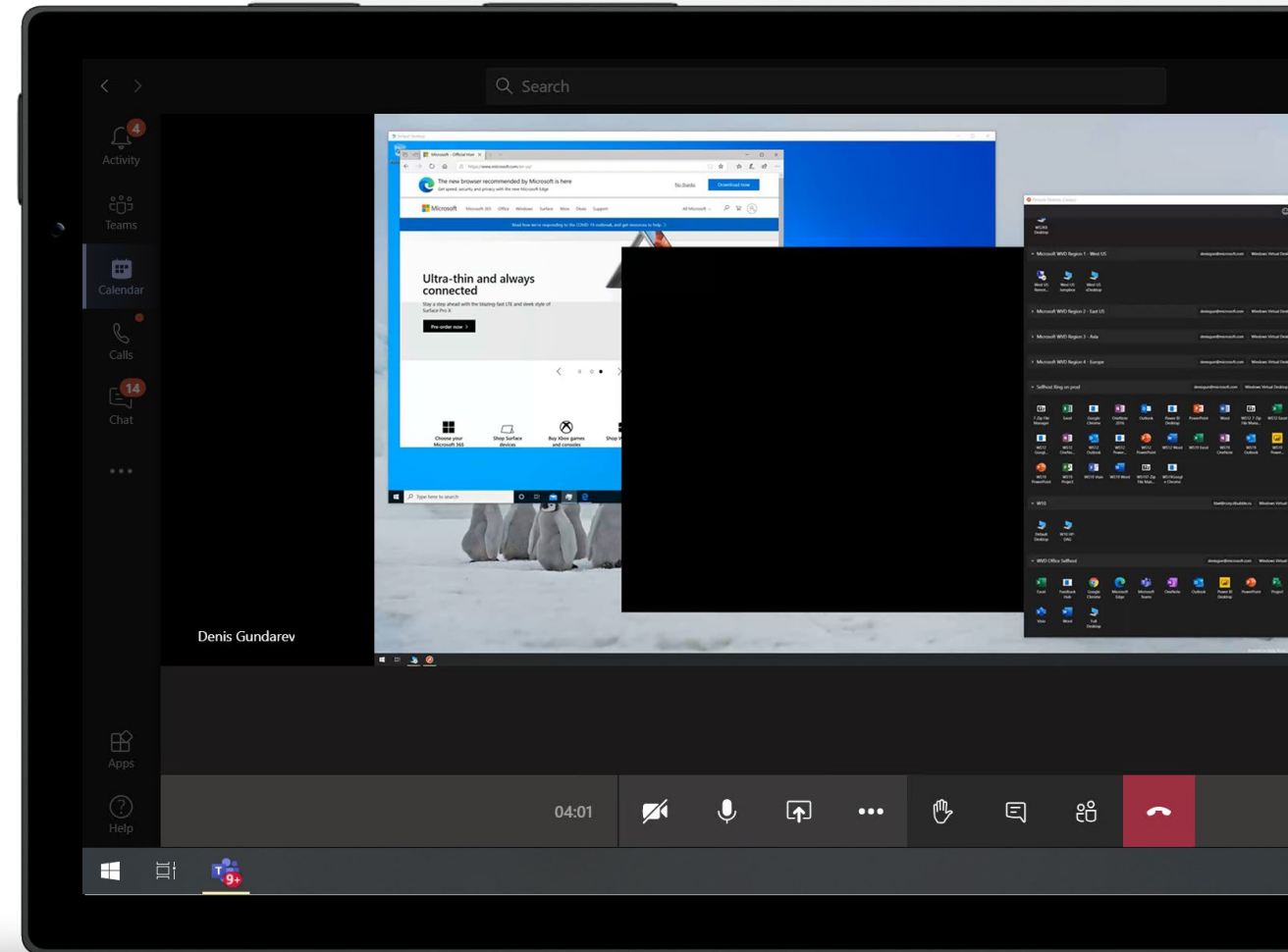
Prevent data exfiltration by limiting clipboard transfer direction to client to session host

Screen capture protection

Disable screen capture for your remote apps and desktop on all the supported Azure Virtual Desktop clients

Watermarking

Enables QR codes which are displayed on the session desktop. Deterrent, preventing sensitive information from being captured and shared.



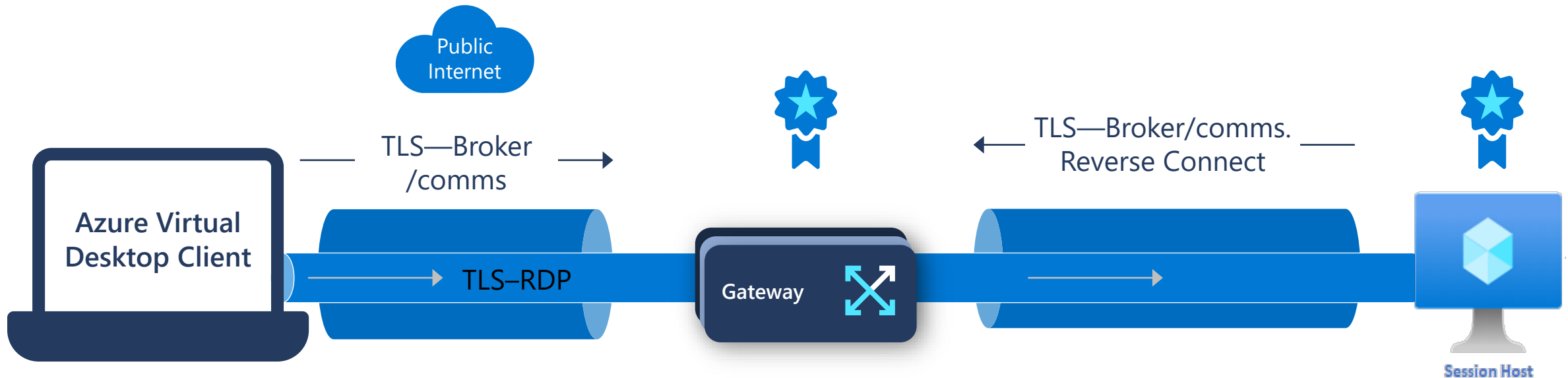


Performance Optimizations

Reverse Connect

Disable all inbound traffic

- ❑ Default connection mechanism
- ❑ No configuration (unless behind firewall)
- ❑ Outbound WebSocket connections from VMs to Broker and Gateway
- ❑ Bidirectional communications between VMs and AVD infrastructure over https (443)
- ❑ No inbound ports need be opened on the VM.



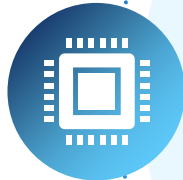
RDP Shortpath

Why UDP?

Improve performance for both
interactive and bulk data transfer
scenarios



UDP doesn't suffer from TCP's
enforcement of its congestion-
avoidance algorithm



UDP does not require the overhead
of communication between the
sender and receiver, and because
no error checking is done, the
sender and recipient can
communicate more quickly



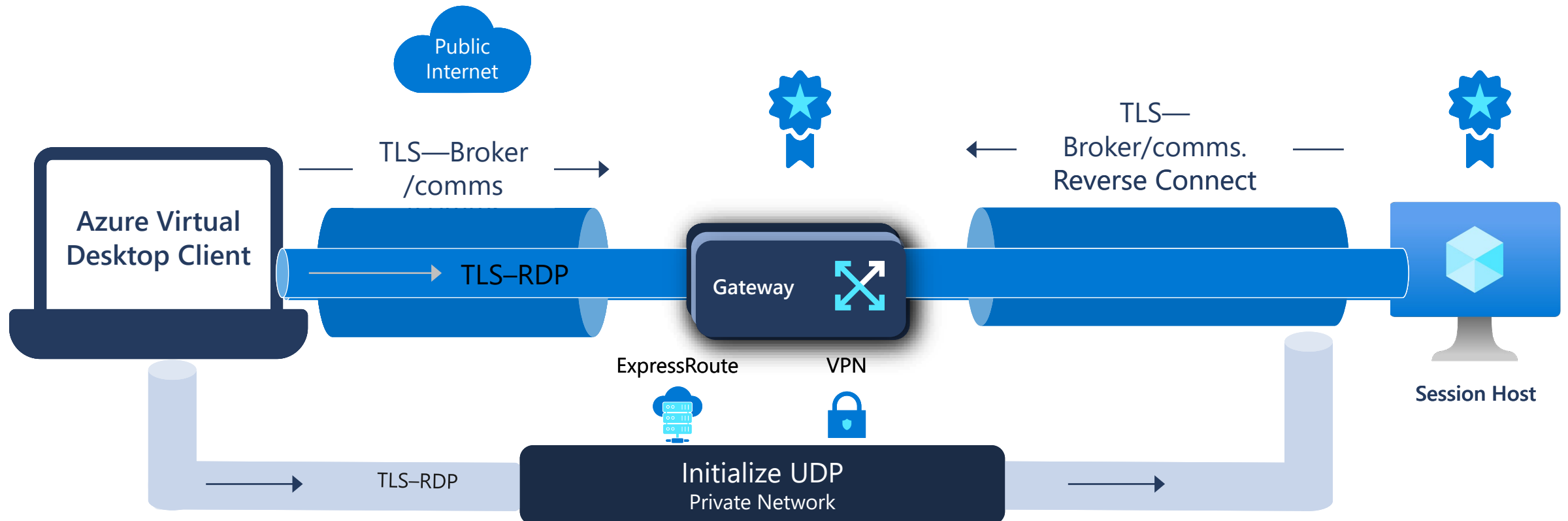
Achieve faster convergence to the
network's bandwidth capacity



RDP Shortpath

for managed networks

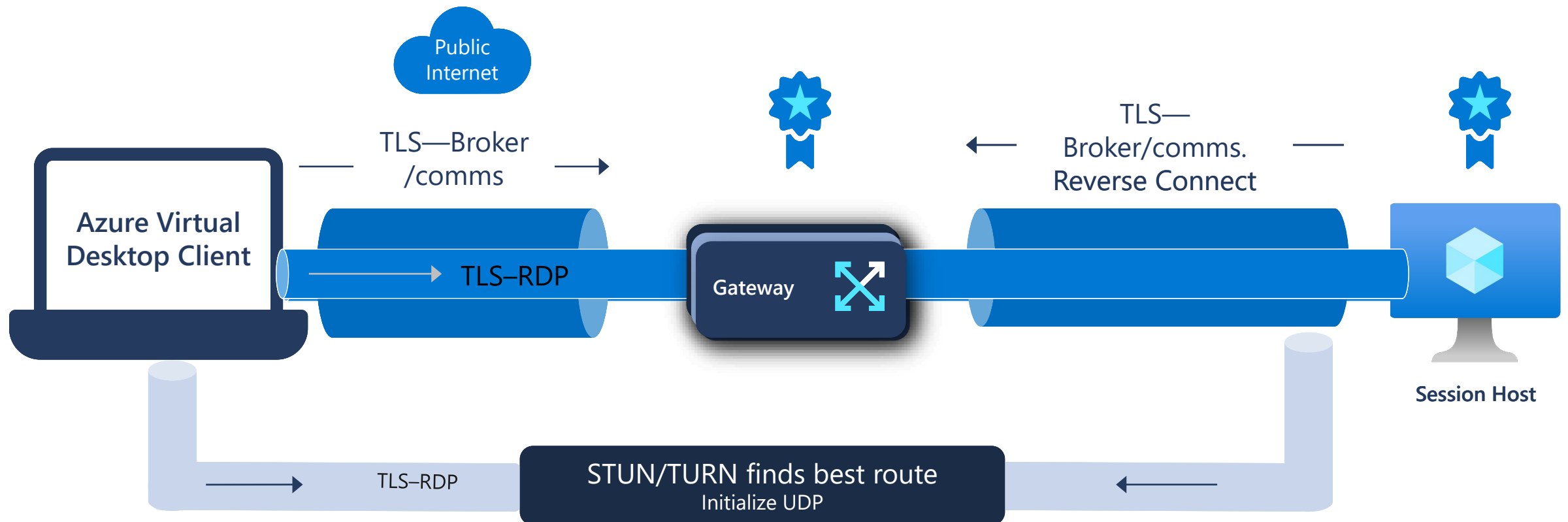
- ❑ Redirects the RDP traffic across your managed network directly and privately to the AVD VMs on your Vnet
- ❑ Requires ER, S2S VPN
- ❑ Reduces the dependency on the AVD gateways, improves the connection's reliability and reduces the round-trip time



RDP Shortpath

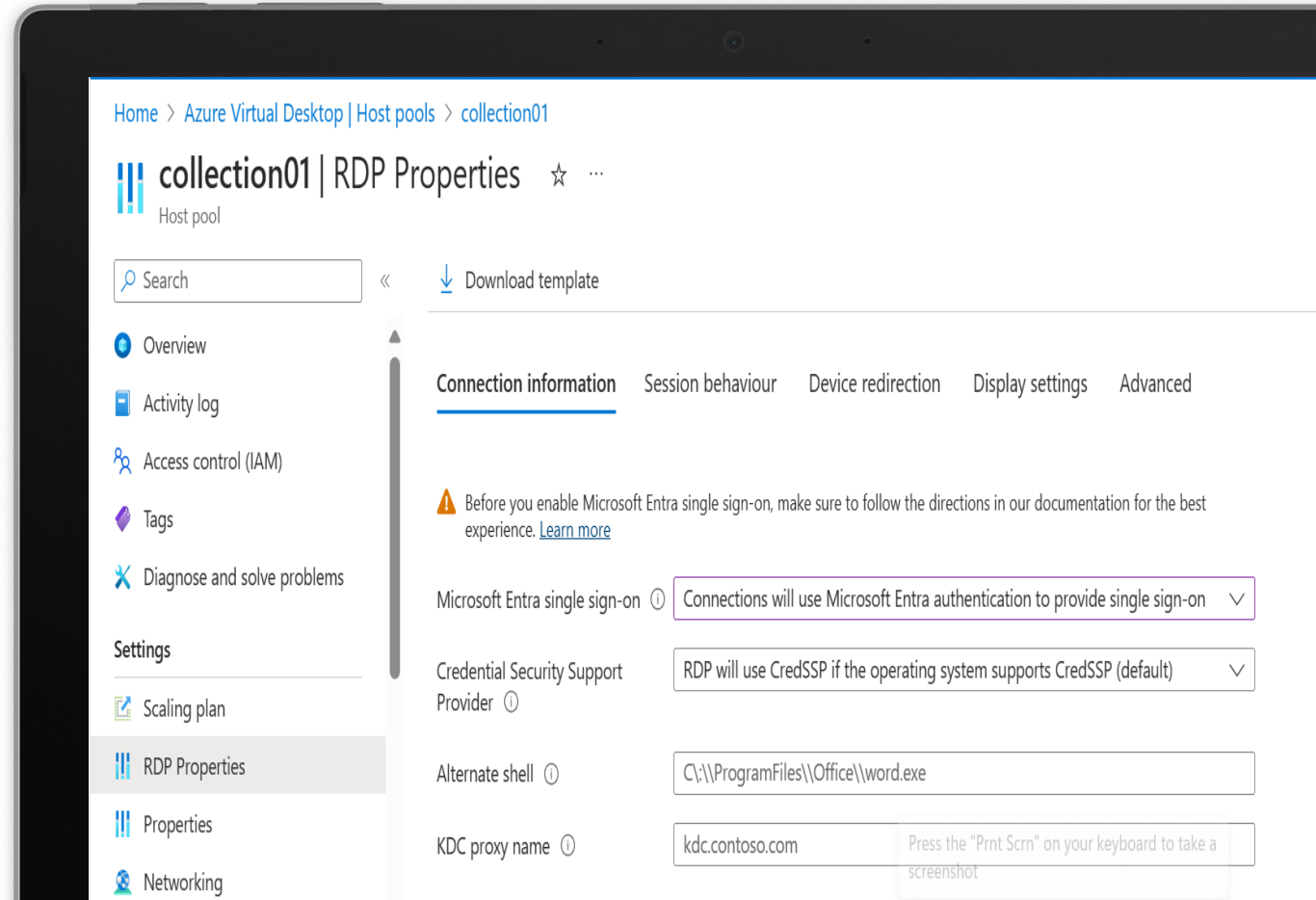
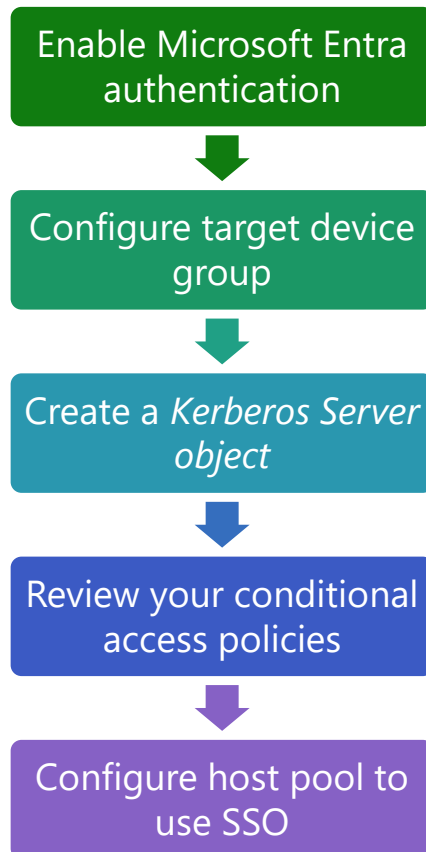
for public networks

- ❑ Redirects just the RDP traffic to use the best connection method possible, enabled by default
- ❑ Avoids the AVD gateways, increases available bandwidth and provides better latency than Reverse Connect
- ❑ During the preview, RDP Shortpath for managed networks is incompatible with RDP Shortpath for public networks



Configure SSO using Microsoft Entra ID

Enable single sign-on (SSO) for Azure Virtual Desktop with Microsoft Entra ID authentication, allowing users to use a Microsoft Entra ID token for password-less authentication and seamless sign-in via third-party identity providers that federate with Microsoft Entra ID.



Password-less Authentication and AVD

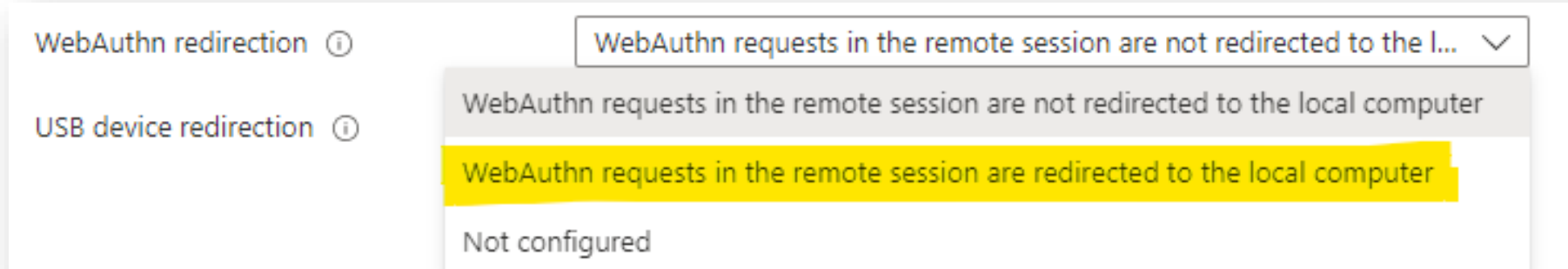
For **cloud service authentication (AVD)**:

- Use any authentication type supported by Entra Id, such as [Windows Hello for Business](#), smart cards, and other [password-less authentication options](#) (for example, FIDO keys)

For the **session host (VM) authentication**:

- The **Windows Desktop client** supports the following authentication methods:
 - Username and password
 - Smart card
 - [Windows Hello for Business certificate trust](#)
 - [Windows Hello for Business key trust with certificates](#)
 - [Entra Id authentication](#) (SSO)

In-session password-less authentication



- Azure Virtual Desktop supports in-session password-less authentication using [Windows Hello for Business](#) or security devices like FIDO keys when using the [Windows Desktop client](#).
- When enabled, all WebAuthn requests in the session are redirected to the local PC. You can use Windows Hello for Business or locally attached security devices to complete the authentication process.
- To access Azure AD resources with Windows Hello for Business or security devices, you must enable the FIDO2 Security Key as an authentication method for your users. To enable this method, follow the steps in [Enable FIDO2 security key method](#).